

Metricity as Licensed Description: The L-metric Boundary and the Realized Physical Interior

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Abstract

Physical theories routinely use metric structure—distance, duration, locality, propagation, curvature, scale, and metric-dependent parameters—as part of their operative descriptive apparatus. This manuscript asks the prior question: under what structural conditions may metric description be used at all?

Actual non-degenerate determinate incidences necessarily instantiate the *Admissibility And Standing Constraint* (AASC) kernel, whether or not they are observed. Qualification for a particular metric use remains answerable to that use’s original conditions. We carry the current AMetric incidence, fibre, selector, and grounding results into an independently specified metric-use problem. Metric structure can be part of its original physical conditions; it cannot supply its own missing qualification merely by being asserted or used.

We retain weak candidate interpretability before target-specific standing and define an L-metric boundary as a locally minimal projection interface where physical metric interpretation is available. From original answers, application domains, joint conditions, and a declared continuation contract, we construct the greatest sound collection of metric-use records and its licensing relation. The central theorem verifies domain qualification, redescription, boundary-role preservation, continuation compatibility, and non-generativity separately. Admissible metric use is licensed interior articulation; the licensing interface generates neither metric structure, standing, nor ontology. Nonempty realization and existence of a minimal interface remain distinct from the construction of the licensing relation.

Physics-bound audits—Lorentzian comparison, zero-point vacuum energy, Standard Model structure and parameters, cosmological transport, and entanglement/no-channel structure—are applications with their own source hypotheses. They are not premises of the licensing theorem. The manuscript introduces no new empirical prediction or replacement physical dynamics. Its proof obligations are original-domain interpretation, semantic coverage, and the displayed preservation arguments; a missing interpretation, omitted required use, or invalid inference remains open to direct criticism.

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1 Introduction: why metricity needs a license

Modern physics is formulated from inside already metric regimes. Relativistic models use intervals, cones, manifolds, and curvature; quantum theories use records, operators, amplitudes, and probabilities tied to measurement contexts; cosmology compares calibrated distances, expansion parameters, and cross-regime observables. None of that practice is rejected here. The prior question is whether metric or metric-dependent predicates may carry standing before the regime in which they are used has itself been licensed.

For a physics reader, the point is not that metric geometry, quantum records, fields, or cosmological parameters are unreal. The point is that their use presupposes a licensed descriptive regime. A Lorentzian interval requires fixed relata; a vacuum-energy scalar requires quotient-stable predictive status; a cross-regime Hubble comparison requires transport closure; an entangled correlation does not by itself supply a controllable channel. The L-metric boundary names the interface at which such descriptors become lawful without becoming primitive.

For actual non-degenerate determinate objecthood, the *Admissibility And Standing Constraint* (AASC) kernel is instantiated by necessity, including at unobserved physical incidences [6]. Its constitutive argument precedes the particular representation and use theorems. The scientific dependency order is

non-degenerate target regime $\Rightarrow$ AASC kernel $\Rightarrow$ original-incidence accountability,

followed by the AMetric determination and grounding results and the present metric-use construction under their stated original-domain hypotheses. Kernel necessity does not by itself determine a metric, prove a physical law, or establish every proposed transfer. The current Structure and Exhaustion results supply complete original-use profiles and exact semantic failure witnesses; ATS supplies the decomposition of interpreted variations; Unus supplies exact interior transport [7, 8, 9, 10, 11]. None of the physics audits is used to establish this chain.

The May bibliography and its earlier companion imports are retained as historical provenance. New current-edition citations identify the source theorem actually invoked in this revision; they do not silently replace older dependencies frozen inside those sources. The proof below also gives the receiving carrier, question, and preservation conditions so that citation identity is not mistaken for theorem application.

1.1 The inversion

The paper fixes the following inversion:

The question is not how metric spacetime generates admissibility. The question is how admissibility licenses metric spacetime description.

This inversion leaves ordinary physics intact inside the physical interior. It denies only the move from interior usefulness to primitive boundary authority. The companion L-metric note states the seed licensing result [3]; the larger AMetric boundary theorem supplies the upstream exclusion package [4]. The present contribution specializes that forced upstream structure to metric description.

1.2 Translation table

The following table fixes the physics-facing reading of the AASC vocabulary used below.

AASC term	Physics-facing reading
Standing	Admissible identity-bearing status of a physical claim, object, comparison, or use inside a fixed scope.
Boundary authority	A role that would authorize admissibility before licensed interior description is available.
Licensed interior	A domain in which physical and mathematical descriptors may be used lawfully.
Realized physical interior	The admitted physical domain in which records, intervals, fields, particles, transport relations, and observables can function as standing-bearing interior structure.
Skin	A variation preserving the complete original content and context; chart or unit changes qualify only on their proved transport laws.
Tensor content	Quotient-stable physical, structural, transport, or witness-bearing content.
Carrier	Structure preserving standing under admissible transformation or comparison.
Scope transport	A source-to-target use requiring an independently justified interpretation and preservation of the target's original conditions.
Undeclared hidden remainder	Standing-bearing direction outside the declared witness, observable, or tensor ledger.
Exhaustion	Proof by eliminating all same-regime alternatives in a declared comparison class.

The terminology is not load-bearing by itself. The load-bearing content is the role package: stable target fixation, identity-preserving reuse, admissible continuation, irreversible commitment, and standing-bearing application. A critic may replace the vocabulary, but must either preserve those functional roles or show that one of them is unnecessary for the target regime.

1.3 Physics-bound audit roadmap

The audits are not premises of the L-metric theorem. They are independent stress tests showing how the theorem classifies distinct uses of metric or physical structure after the boundary theorem has already been stated.

Physics-bound audit	Standard issue	L-metric classification
Lorentzian interval	Metric comparison of events	Interval comparison requires standing-positive relata.

Physics-bound audit	Standard issue	L-metric classification
Zero-point vacuum energy	Vacuum-energy scalar and cosmological constant	Non-invariant offsets fail standalone prediction status.
Standard Model tuple	Structure versus empirical parameter table	Structural licensing does not close every metric or numerical parameter.
Hubble transport closure	Comparing early/late H_0 determinations	Same-parameter comparison requires transport closure.
Entanglement/no-channel	Nonlocal correlation versus signaling	Joint standing does not supply a controllable boundary channel.

The audits do not mutually certify one another. The L-metric theorem supplies the classification rule; each audit applies that rule to a separate physics-facing structure. Failure or revision of one audit would affect that audit, not the central licensing theorem. The audits are schematic classification cases, not full treatments of the underlying physical theories.

1.4 Position relative to standard physics

The manuscript is adjacent to familiar issues in physics foundations: spacetime structure in relativity, operational no-signaling in quantum theory, observables, renormalization-scheme dependence in QFT, and cross-regime parameter comparison in cosmology. It is not a replacement for Lorentzian geometry, Bell-type locality analysis, quantum no-signaling theorems, renormalized stress-tensor theory, or cosmological model comparison. It supplies a constraint-theoretic classification of when their metric or structural descriptors are licensed as standing-bearing physical description. External literature is used where needed to anchor the worked audits, especially the Lorentzian and zero-point examples [46, 45, 42, 43, 44, 47, 48, 49, 51, 52, 53, 54, 56, 57, 60].

1.5 Representation is not unreality

Licensed interior description is not unreality. A field, particle, metric, record, gauge structure, or structural tuple may be physically real as a role-stable interior object. What is denied is primitive boundary status: the ability of such an object, before licensing, to authorize the standing of the regime in which it is used.

1.6 Claims and nonclaims

Claims. Metric descriptors cannot supply primitive authority for their own missing qualification. L-metric licensing is constructed from original qualified uses and their preservation conditions. Cross-regime comparison requires its actual transport theorem. Quotient failure blocks a claimed scalar presentation-independent prediction. Joint standing alone supplies no additional boundary-controlled communication channel.

Nonclaims. The paper does not derive GR or quantum dynamics, does not derive the full Standard Model parameter table, does not solve the cosmological-constant problem, does not adjudicate unrestricted Λ CDM model selection, does not derive Hilbert-space no-signaling from quantum dynamics, and does not assert global uniqueness of all possible L-metric interfaces.

1.7 Manuscript architecture

The manuscript is organized in three parts. Part I states the boundary theorem: the formal core, the constraint-formalism proof mode, the AMetric exclusion discipline, the L-metric definition, the licensing/no-backflow results, and locality as a licensed-interior corollary. Part II gives independent physics-bound audits: Lorentzian metric comparison, zero-point quotient descent, structural/parametric separation in the Standard Model tuple, Hubble transport closure, and entanglement/no-channel classification. Part III states consequences, limits, dependency structure, and refutation criteria.

Part I

Boundary theorem

2 Self-contained formal core

This section states the formal vocabulary used by the manuscript. The results concern independently specified original uses: their answers, applicability, and promised continuations. The proofs construct a licensing relation and identify exact semantic failures of proposed alternatives. Role names organize those failures after their witnesses have been established; the names do not establish exhaustion.

2.1 Fixed-domain evaluation package

Definition 2.1 (Background evaluation package). A background evaluation package is a tuple

$$D_{\text{bg}} = (\text{Act}_D, \text{Eval}_D, \text{Red}_D, \equiv_D, \text{Cont}_D, B_D^{\text{place}}),$$

where:

- (i) Act_D is the class of acts, states, reports, or constructions considered under the same domain typing;
- (ii) $\text{Eval}_D \subseteq \text{Act}_D$ is the evaluated fragment;
- (iii) $\text{Red}_D \subseteq \text{Act}_D \times \text{Act}_D$ is admissible redescription;
- (iv) $\equiv_D$ is identity under admissible redescription and retained continuation;
- (v) Cont_D is the finite-history continuation structure for same-domain retained histories; and
- (vi) B_D^{place} is the boundary-place/interface type at which standing-bearing admission is evaluated, before any disputed role such as generativity, metricity, non-transmissivity, or non-generativity is assigned.

Definition 2.2 (Standing map over a background). A standing map over D_{bg} is a partial map

$$L_D : \text{Act}_D \rightarrow \{0, 1\}, \quad \text{dom}(L_D) = \text{Eval}_D.$$

The standing-positive interior is

$$\text{Int}_D^+ := \{a \in \text{Eval}_D : L_D(a) = 1\}.$$

Undefinedness is not a third standing value. It is absence of evaluation.

Original conditions before the standing report. For each evaluated act a , the domain independently identifies its bearer, original occurrence, complete use question, and substantive qualification condition $Q_D(a)$. These are the conditions of the physical or mathematical use itself, not conditions selected to make a proposed classifier correct. A report of a measured interval, for example, concerns the identified events and the stated metric, calibration, and comparison conditions. An actual inscription of a false interval report and a correct report of the interval answer different qualification questions.

The faithful standing map for this specified question records

$$L_D(a) = 1 \iff Q_D(a). \quad (1)$$

This is a semantic accountability requirement on a standing report. The original incidence fixes Q_D before its characteristic function is written. A competing map is admitted for examination and must establish its own correctness against those same conditions. In classical logic the faithful Boolean map exists and is unique on the evaluated fragment: at each act its value records $Q_D(a)$ or its negation. Agreement between classifiers is a conclusion. A partial practical checker can be sound without deciding every evaluated question; its silence is not 0. This is the original-incidence construction of the Kernel, Definition 2.5 and Proposition 2.6 [6].

Definition 2.3 (Non-degenerate target regime). A non-degenerate target regime contains an actual identity-bearing object or incidence whose relata and original conditions distinguish retaining that incidence from substituting another. It need not be observed, assessed by an agent, or actually reused. Repeated identification and physical-description practices are applications of this target class. A bare uninterpreted variable supplies no physical realization merely by being written; an actual inscription nevertheless has its own determinate incidence, distinct from the object or physical use it purports to describe.

Lemma 2.4 (Standing-bearing physical-description regimes instantiate the target class). *A physical-description regime concerning actual determinate objects, events, records, or comparison incidences instantiates the non-degenerate target class. Its actual incidences necessarily perform the Kernel's Reference, Standing, Irreversibility, and Admissibility work. Qualification of a particular proposed metric use remains answerable to that use's original conditions.*

Construction witness. Begin with the actual incidence, not a declaration that its description is adequate. Substituting different relata loses its subject; failing its constitutive conditions loses the specified relation; satisfying different conditions in incompatible arrangements loses their joint realization; substituting a later occurrence loses its original attribution. Retaining the incidence therefore requires the work named Reference, Standing, Admissibility, and Irreversibility respectively. These losses are the constitutive argument of the current Kernel, Proposition 2.4; its Theorem 3.1 reconstructs the four functions from those commitments [6]. The functions may overlap or be distributed. Their necessity requires neither an observer nor a separate checking event. The positive qualification of an inscription for its own occurrence does not establish the admissibility of the metric claim made with it: that second use must satisfy its own Q_D . $\square$

Definition 2.5 (Background same-scope and standing-preserving same-scope). A construction remains *background same-scope* for D if it preserves Act_D , Eval_D , Red_D , $\equiv_D$, Cont_D , and B_D^{place} as boundary-place/interface type. It does not automatically preserve the boundary's disputed role as non-generative, non-metric, non-transmissive, or non-authorizing. A background same-scope construction may therefore propose a competing standing map L'_D or a competing role assignment over the same background.

A construction remains *standing-preserving same-scope* if it is background same-scope and also preserves L_D up to extensional equality on evaluated acts. Standing-changing challenges are evaluated against background same-scope, not defined away by standing preservation.

For a claim to answer the *same original question*, the bearer, occurrence, use, and substantive conditions interpreting this background must also be retained. This fixes meaning, not a preferred verdict. A new map over the same displayed sets can be tested against the original conditions; disagreement alone is not evidence that its scope changed.

Definition 2.6 (Boundary authority). A datum q has boundary authority over a background D_{bg} if varying q while preserving the background changes at least one of:

- (a) a candidate boundary standing classification $L_D(a)$ for some evaluated act a ;
- (b) whether a same-domain retained history is coherent under Cont_D ;
- (c) which boundary outcome is selected among alternatives; or
- (d) which metric/interior descriptors are admitted as standing-preserving descriptions.

This definition records an alleged effect; it does not establish that the effect is warranted. *Primitive boundary authority* means that the datum is offered as the source of its own target qualification, or as a replacement for the unchanged original conditions without an independently warranted incidence or transfer. An original metric condition can genuinely enter Q_D ; a sound test can establish it. Neither is primitive self-authorization. If q changes none of the listed answers, it is inert for these questions only; complete representational inertness also requires preservation of every other required original answer and its definedness.

Definition 2.7 (Candidate application class). For a descriptor m , let $X_D(m)$ be the syntactic or formal candidate application class on which m may be written before physical interpretability has been established. Elements of $X_D(m)$ need not yet be standing-positive. This class prevents the licensing theorem from assuming the conclusion that all metric use is already interior.

The candidate is a proposed *application*, including its intended use question, rather than an untyped physical bearer. When its interpretation is established, write $a = \iota_m(x) \in \text{Eval}_D$ for that original use. For ordinary single-domain uses we identify x with a . A composite requires its joint-use interpretation; separate qualifications of its parts do not establish the composite's qualification.

2.2 Metric descriptor classes

The manuscript does not treat all physical quantities as one homogeneous class. It uses a stratified descriptor family. Each class is metric-dependent in a different way, but the licensing theorem applies through the same fixed-domain route-exhaustion argument.

Definition 2.8 (Stratified metric descriptor family). For a fixed physical domain D , let

$$\mathcal{M}_D = \mathcal{M}_D^{\text{dist}} \sqcup \mathcal{M}_D^{\text{top}} \sqcup \mathcal{M}_D^{\text{causal}} \sqcup \mathcal{M}_D^{\text{dyn}} \sqcup \mathcal{M}_D^{\text{obs}} \sqcup \mathcal{M}_D^{\text{const}}.$$

The components are:

- (i) $\mathcal{M}_D^{\text{dist}}$: distance, duration, length, norm, scale, interval, volume, rate;
- (ii) $\mathcal{M}_D^{\text{top}}$: neighborhood, support, boundary, horizon, global/local connectedness when used as physical articulation;
- (iii) $\mathcal{M}_D^{\text{causal}}$: causal order, light-cone structure, propagation admissibility, signal constraints;

- (iv) $\mathcal{M}_D^{\text{dyn}}$: equations, transport rules, curvature/dynamical relations, field evolution laws;
- (v) $\mathcal{M}_D^{\text{obs}}$: reconstructed observables, calibrated records, luminosity distances, sound-horizon inferences, data-model comparison quantities;
- (vi) $\mathcal{M}_D^{\text{const}}$: dimensionful presentations of constants, unit-dependent parameter expressions, and dimensionless ratios proposed as invariant content.

Definition 2.9 (Weak interpretability). A candidate use (x, m) with $x \in X_D(m)$ has *weak interpretability* when it is syntactically, model-theoretically, or computationally well-formed in a candidate presentation. Weak interpretability may be available before physical standing has been established; it is not yet physical-standing interpretability.

Definition 2.10 (Physical-standing interpretability). A candidate use (x, m) has *physical-standing interpretability* when its relata, records, calibration standards, unit conventions, comparison rules, witness ledger, and continuation role have their target-fixed interpretations in the target domain. A witness-preserving admissible use must actually satisfy the original conditions for the claims and continuations it undertakes. Interpretability alone does not make a false physical assertion true. The licensing theorem proves that an *admissible* such use has $\iota_m(x) \in \text{Int}_D^+$, including the independently assessed joint-use act when the application is composite. Neither $L_D(x) = 1$ nor membership in Lic_D is a premise of weak interpretability.

Definition 2.11 (L-metric dependency condition). A metric descriptor $m \in \mathcal{M}_D$ satisfies the L-metric dependency condition if it may be written on a candidate class $X_D(m)$ and may possess weak interpretability there, but its physical-standing interpretability is not assumed. The theorem below derives the consequence

$$\text{dom}_D^{\text{phys}}(m) \subseteq \text{Int}_D^+$$

for admissible fixed-domain metric uses; it is not built into the definition. Here $\text{dom}_D^{\text{phys}}(m)$ denotes the original-use acts of admissible physically interpreted applications, not every meaningful formula or inscription of a claim. Candidate well-formedness, established interpretation, and actual qualification are separate questions.

2.3 Projection order and licensing

Definition 2.12 (Projection order). Let P_1 and P_2 be projection interfaces over the same background D_{bg} . Write

$$P_1 \preceq_D P_2$$

when standing-preserving use of P_2 requires P_1 as a prerequisite under Cont_D and under admissible redescription. This is a dependency order, not a temporal order. The manuscript uses minimality relative to this order only locally; it does not prove that a unique global minimal metric interface exists across all possible backgrounds.

Definition 2.13 (Descriptor equivalence). For $m, m' \in \mathcal{M}_D$, write

$$m \sim_D^{\mathcal{M}} m'$$

iff, under independently justified identifications of their input and output types, they have the same complete original-use profile: required comparisons, record relations, continuation and joint-domain conditions, observables, and calibrated responses, including definedness. These uses are fixed before constructing Lic_D . Coordinate, gauge, unit, chart, and definitional changes belong to this equivalence when their transport laws preserve those meanings. Raw tensor components need not be equal under a change of coordinates.

Definition 2.14 (Constrained L-metric licensing relation). The L-metric licensing relation is a partial relation

$$\text{Lic}_D \subseteq \text{Int}_D^+ \times \mathcal{M}_D.$$

The assertion $(a, m) \in \text{Lic}_D$ means that standing-positive content a may be articulated using descriptor m . The relation is constrained, not arbitrary. It must satisfy:

- (L1) *domain constraint*: the first coordinate is standing-positive;
- (L2) *redescription invariance*: if $a \text{ Red}_D a'$ and $(a, m) \in \text{Lic}_D$, then there exists $m' \sim_D^{\mathcal{M}} m$ such that $(a', m') \in \text{Lic}_D$;
- (L3) *no boundary mutation*: licensed use does not alter the role or constraint-status assigned to the boundary-place B_D^{place} ;
- (L4) *continuation compatibility*: licensed use preserves Cont_D or changes only declared interior continuation structure;
- (L5) *non-generativity*: the original qualification of a grounds its licensing; membership can certify that qualification but does not create it or replace its original conditions.

Definition 2.15 (Boundary mutation). Boundary mutation is an alleged change in the role, function, or constraint-status assigned to B_D^{place} through a redescription of interior structure. It is not automatically a change of background, since background same-scope comparison leaves the disputed role open. Its admissibility must be decided against the independently fixed original question and the AMetric incidence, fibre, or grounding conditions. Renaming a role does not establish that its function changed.

Definition 2.16 (Carrier, selector, and scope-transport failures). The common failure modes are:

- (i) a *carrier* failure, where a metric success, score, probability, invariant, or quotient coordinate substitutes for an unestablished original-use warrant;
- (ii) a *selector* failure, where a proposed choice lacks the eligible fixed point or original-incidence warrant required by its claimed interpretation;
- (iii) an *operator* failure, where a proposed application asserts its own domain or license without establishing the original conditions;
- (iv) a *scope-transport* failure, where an interior or source-domain structure is moved into the boundary or target domain without independent target gating; and
- (v) a *boundary-mutation* failure, where the boundary role changes with the interior presentation.

2.4 Standing and empirical warrant

Definition 2.17 (Empirical warrant versus admissibility standing). Empirical warrant is evidential support for using, comparing, ranking, or continuing a physical model inside a scientific practice. Admissibility standing is the fixed-domain authorization status $L_D(a) \in \{0, 1\}$ on the evaluated fragment. Empirical warrant may affect theory choice inside the interior. It is not identical to primitive boundary authorization.

Bayesian confirmation, posterior credence increase, successful retrodiction, and calibration fit are interior epistemic updates over already meaningful hypotheses. The no-backflow theorem concerns a different operation: using those successes as pre-metric boundary authorization for the hypotheses' own standing.

2.5 Minimal proof target

Proposition 2.18 (Minimal L-metric proof target). *For the independently interpreted original metric-use task, construct Lic_D and prove that every admissible physically interpreted use belongs to it and satisfies (L1)–(L5). Establish non-self-authorization by the actual AMetric incidence, fibre, stabilizer, and grounding arguments. Any claimed failure of the original-use requirements must have its corresponding semantic witness; naming a carrier, selector, or operator does not suffice.*

This proposition is not yet the theorem. It is the target against which the subsequent sections are organized.

3 AASC as a constraint formalism

AASC is a constraint formalism. Its necessity claim concerns actual determinate realization; its application theorems concern the preservation of independently specified original conditions. Exhaustion requires both coverage of the original task and a proof of the alternatives for its realization. A finite taxonomy, a registry entry, or the absence of a discovered counterexample supplies neither proof.

Principle 3.1 (Kernel instantiation is forced, not stipulated). *Every actual non-degenerate determinate incidence necessarily performs the Kernel work established in Lemma 2.4. A failed trial or false report remains a determinate incidence; its qualification for a different claimed use can fail. Particular physical laws, metric relations, and transfer conditions remain substantive facts of their original domains.*

Definition 3.2 (Exhaustion classifier). For a fixed original task over D_{bg} , let $\mathcal{V}_D$ contain the interpreted proposals for its answers, representations, and continuations, including proposals whose standing is disputed. An exhaustion classifier is a multivalued diagnostic relation

$$\mathcal{E}_D \subseteq \mathcal{V}_D \times \mathcal{R}_D.$$

The dispositions are equivalence, scope change, selector, carrier, operator, repair, hidden remainder, mutation, skin, and failure. These familiar names identify proposed mechanisms. A failure disposition is justified only by an original semantic defect: a wrong answer, lost required distinction, unsafe promised use, or invalid offered support. A lawful selector, metric, operator, or new constructor is not a failure merely because of its name. The relation need not be effective, exclusive, or ordered by priority.

Definition 3.3 (Closed fixed-domain exhaustion). An exhaustion theorem is closed for an original task when its interpreted requirement family covers that task and every proposal either meets its requirements or has a proved defect relative to them. Soundness alone does not establish complete coverage. An undisclosed interpretation or missing proof is recorded as unestablished; it is not a proved negative physical qualification.

Semantic coverage construction. Let U_D be the independently specified set of original questions and required uses, with evaluated presentations E_D . Each question u has its original result relation $G_u \subseteq E_D \times V_u$, possibly partial. When the required result is functional, totalize it by a distinct absence tag and form the complete profile

$$P_D(e) = (o_u(e))_{u \in U_D}, \quad o_u(e) \in V_u \sqcup \{\perp_u\}. \quad (2)$$

For a relational problem with several permitted results, retain the Boolean membership questions for its original graph instead of selecting a result. Identity indexing retains every original question, its definedness, and its interpretation. Thus this semantic profile covers the specified task without a finite list of implementation styles. It is not an effective oracle or a demonstration that an arbitrary physics inventory is complete. A smaller list or generated language requires a proof that each original use has its correctly interpreted coordinate or construction. Original joint domains and any required readiness, historical, or eventual-outcome questions must be included, not inferred from a present qualification bit.

Here are the exact alternatives. An asserted result graph K_u is sound when $K_u \subseteq G_u$; complete coverage on the original required domain M_u also needs $G_u \cap (M_u \times V_u) \subseteq K_u$. Violation has a surplus or omitted original record. A proposed encoding $h : E_D \rightarrow H$ retains the complete profile exactly when

$$h(e) = h(e') \implies P_D(e) = P_D(e'). \quad (3)$$

Necessity follows because a reader at one code has one profile. Conversely, define the reader on $h(E_D)$ by the common profile of its fibre; the implication makes it well defined and surjectivity onto the image makes it unique. Failure supplies an equal-code pair and a required coordinate on which the original answers differ. For a declared continuation contract, Section 6 supplies the independent node, edge, and safe-interior construction. A false conclusion in a well-founded interpreted certificate has a false leaf or a minimal false inference whose premises are true. A missing or defective proof of a true conclusion must be distinguished from a false conclusion.

These arguments instantiate Structure, Lemma 8.4 and Theorem 8.6, and Exhaustion, Theorems 3.2 and 5.9 [8, 9]. Their proofs run from original requirements to witnesses; diagnostic mechanism names are assigned afterwards. A faithful recoding cannot eliminate an original graph difference, split an unchanged colliding input fibre, or make an unsafe original path safe. Correcting the actual defect can establish a new faithful realization. No finite audit is a premise of this semantic result.

Principle 3.4 (Theorem status in a constraint formalism). *The theorem labels mark claims with stated original-domain interpretations and mathematical arguments. Necessity of the Kernel does not automatically discharge an application's coverage, transport, positive-realization, or physical-law obligations. Once those interpretations are established, the original-graph, fibre, continuation, and support alternatives cover arbitrary implementations of the same specified task.*

Remark 3.5 (Generator demands). A request for a physical mechanism is separate from this paper's licensing result. A proposed constructor is nevertheless a legitimate mathematical challenge when it claims to establish the original use. Its input domain, result incidence, and warrant must be checked. A sound constructor can supply qualification for a new output; it cannot replace the failed conditions of an unchanged original occurrence by naming that output a repair.

Remark 3.6 (Coverage, not priority). A defect can have several descriptions, such as lost information and unwarranted transport. Exhaustion does not require one preferred name. A relevant original use missing from the alleged complete family defeats its coverage claim even before a counterexample to the conditional mathematical theorem has been constructed.

Remark 3.7 (Undeclared hidden remainders). An unobserved physical direction is not forbidden by being unobserved. If erasing it changes a required original answer or application domain, it belongs in the complete profile or must be recovered by a proved decoder. If it supplies a new task, that task requires its own interpretation. Information absent from a proposed representation is not thereby absent from the original incidence.

4 AMetric exclusion: why boundary authority is non-metric

The upstream result concerns the source of qualification authority, not a prohibition on writing a metric, using an operator, or discovering a physical relation. We use the constructed Axiom 6 of the current AMetric manuscript, Theorem 7.5, with its exact source and receiving conditions [7]. The May companion remains part of the historical provenance [4].

4.1 The retained-content problem

Definition 4.1 (Pre-fixational stage). A proposal interface is pre-fixational for a target discrimination when that discrimination has not been fixed by the interface's independently warranted retained content. The proposal record itself can be a determinate object. An original physical incidence can also have a determinate answer that the proposal does not retain or that an investigator does not yet know. Pre-fixation is relative to the particular target claim and input, not a kernel-free phase of physical existence.

Lemma 4.2 (No unsupported metric discrimination before fixation). *Let $p : Z \rightarrow X$ retain exactly the original proposal data from richer candidate situations, and let $w : Z \rightarrow V$ be the required metric answer, with output types compared by their independently justified transport. A source-only answer reproducing w exists on $p(Z)$ if and only if*

$$p(z) = p(z') \implies w(z) = w(z').$$

Consequently, an unretained metric distinction cannot be supplied by a function of those unchanged retained data alone. Original-incidence warrant for the answer is a further requirement.

Construction witness. A factorization $w = \bar{w}p$ makes equal retained inputs have equal answers. Conversely, the displayed condition defines $\bar{w}(x)$ as the common answer on the nonempty fibre over x ; it is well defined and unique on $p(Z)$. A discordant pair refutes every source-only decoder, independently of its implementation. A total claim on X additionally requires $p(Z) = X$ or a separately justified extension. This is AMetric, Theorem 7.2. If a separately warranted metric relation was already part of the complete original problem, p must retain it or the proposed source-only claim fails this test. Lack of a chosen representative alone does not erase a relation. $\square$

Lemma 4.3 (Exact representative-selection obstruction). *Let a group G act on the actual retained inputs X and eligible outputs Y , and let $F(x) \subseteq Y$ be the independently specified candidate family with $F(gx) = gF(x)$. An eligible equivariant selector s must satisfy*

$$s(x) \in F(x) \cap \text{Fix}_Y(G_x), \quad G_x = \{g : gx = x\}.$$

An empty intersection at one input rules out such a selector. If every intersection is a singleton, it defines the unique eligible equivariant selector.

Construction witness. For $g \in G_x$, equivariance gives $gs(x) = s(gx) = s(x)$; eligibility gives the intersection. In the singleton case let $s(x)$ be its unique member. The intersections transport by g , so uniqueness gives $s(gx) = gs(x)$ and also uniqueness of the selector. These are the relevant clauses of AMetric, Theorems 4.6 and 5.4. Existence of merely nonempty intersections does not, without further conditions, choose across arbitrary orbits. Eligibility and the original-incidence meaning of the selected value remain substantive. $\square$

When equality is the entire authoritative atom structure on a nonempty carrier P , its lawful group is $\text{Sym}(P)$. An unmarked nonsingleton eligible subset has no fixed eligible atom under its set stabilizer. Scalar tuple observations factor through equality patterns. An invariant metric on P can nevertheless have the discrete form $d(x, y) = c$ for $x \neq y$, with $c > 0$; equality alone supplies no nontrivial distance geometry or preferred scale. These are AMetric, Theorem 6.3, Proposition 6.5, and Corollary 6.8. The equality-only premise must be established. An unmarked four-cycle has an invariant graph metric although no vertex is preferred; its adjacency is additional original structure. Tuple entries and singleton candidates can also provide lawful equivariant selections. None of these cases is excluded by a no-selector slogan.

Lemma 4.4 (Boundary-authority normal form). *An interpreted proposal for boundary authority must answer the original questions and use obligations in Section 3. It therefore either supplies warranted original answers and the required coverage, or has an original-result, information, continuation, or offered-support defect. Familiar routes locate possible defects as follows:*

- (N1) *a competing standing map is compared pointwise with the original Q_D ;*
- (N2) *a proposed selector is tested on the actual eligible family and lawful action;*
- (N3) *an operation application is tested on its original domain, result, and warrant;*
- (N4) *a transfer is tested against the target incidence and claimed target-use coverage;*
- (N5) *a role change is compared with the independently fixed original question and context;*
- (N6) *a structural condition is tested for its original incidence and retained information; or*
- (N7) *an effect preserving the complete original profile is inert for those original answers.*

The names may overlap. They do not themselves prove that a named construction fails.

Construction witness. Every original requirement has its interpreted coordinate by (2). Comparing the asserted and original result graphs supplies surplus or omission; a lost required coordinate has the fibre witness (3); an unsafe promised continuation has a failed original node or edge; and a false supported conclusion has a false premise or invalid interpreted inference. These alternatives are the semantic exhaustion of Structure, Theorem 8.6, and Exhaustion, Theorem 5.9. The listed routes describe where those witnesses arise. A competing map is not dismissed merely for competing, and an operation is not dismissed merely for occurring. A correctly interpreted proposal satisfying all the original requirements is a faithful realization. An alleged missing original requirement calls for a coverage check, not assignment to a residual failure name. $\square$

Lemma 4.5 (Structural conditions do not authorize themselves). *A metric-like structural condition can govern a physical use when it belongs to that use's independently specified original conditions and its application is warranted. If it is instead offered as a source-only discriminator, it must pass the fibre test; if its license is offered only through mutually presupposing applications, it needs an independent grounded entrance. Its asserted status as structural, canonical, or metric-free supplies neither warrant.*

Construction witness. The first case is original-incidence qualification, not an extra source of authority. In the second, a discordant retained-data fibre excludes the claimed decoder by the first lemma. For the third, let B be independently grounded judgments and $\mathcal{R}$ independently sound rules on a judgment set J . Let C be the proposed license family. If $B \cap C = \emptyset$ and every rule concluding in C has a premise in C , then $J \setminus C$ is a rule-closed superset of B . The least grounded rule closure is contained in it and contains no member of C (AMetric, Theorem 7.4). An independently warranted

test or constructor theorem can provide an entrance and then this obstruction does not apply. A missing warrant is not evidence that the underlying physical condition is false. $\square$

Theorem 4.6 (AMetric exclusion theorem). *For an unchanged original qualification question in a non-degenerate determinate regime, no metric, temporal, probabilistic, causal, modal, topological, ordering, selector, index, structural-constraint, or operator structure is a primitive source of its own target-specific qualification. A claimed source-only answer must be determined by the actual retained data and independently warranted for that original question. Additional fixing data require their own incidence and warranted connection to the answer; an unseeded family of presupposed licenses cannot supply that connection.*

Construction witness. Fix the bearer, occurrence, complete question, and original conditions before the proposed authorizer. The faithful qualification reports have the unique value in (1). A parameter can therefore neither add a third qualification value nor make two independently warranted answers to that same question disagree. This does not forbid parameters that retain genuine original distinctions.

For a richer proposed authorizer, specify the actual forgetting map p and target answer w . Either w is constant on the fibres, giving the unique source-level determination, or a discordant pair excludes it. In the first case original warrant is still required; in the second, warranted additional data change what is available to the argument. An allegedly primitive selected output additionally obeys the stabilizer criterion when its interpretation requires equivariance. An operation offered as its own first license is subject to the grounded-closure argument in the preceding lemma. Thus the authority must come from original-incidence warrant or a warranted refinement and transfer, not the asserted authority of the proposed description itself. These are exactly the branches of constructed Axiom 6, AMetric Theorem 7.5 and its export discipline in Section 7.6. Their force comes from the displayed incidence, fibre, stabilizer, and grounding arguments, not from the mechanism names. $\square$

Corollary 4.7 (No primitive self-authorizing metricity). *Metricity cannot supply the missing qualification for its own use merely by being asserted, computed, or made part of a proposed model. Admissible physical metric use must exhibit its original qualification and preservation conditions.*

Remark 4.8 (Simultaneous interior construction). A physical theory may introduce relata and metric structure jointly, and a sound construction can establish a new qualified object. A Lorentzian manifold and metric can be specified together. Their original joint conditions and warranted interpretation do the licensing work. No temporally earlier checker is required; the construction cannot replace that work with its own asserted admissibility.

Remark 4.9 (Scope of the derivation). The exclusion concerns constitutive dependence and the unchanged original question. It does not assert that no physical process, geometric condition, metric map, or invariant relation can exist before an investigator has identified it. Kernel governance is necessary for those actual incidences as well.

5 The L-metric boundary defined weakly

The definition of an L-metric boundary must not smuggle in the licensing theorem. We therefore define it only as a minimal projection interface at which metric descriptors may become physically interpretable. Unless a uniqueness lemma is proved for a given background, the manuscript speaks of an L-metric boundary up to the local projection order, not a globally unique boundary object. The restriction that they are interior-only will be proved in the next section.

Definition 5.1 (Metric projection interface). A metric projection interface P over a background D_{bg} is a projection interface at which at least one descriptor $m \in \mathcal{M}_D$ has a nonempty physically interpretable domain of use, provided the use is stable under admissible redescription and compatible with same-domain continuation.

Definition 5.2 (L-metric boundary). An L-metric boundary B_D^L is a $\preceq_D$ -minimal metric projection interface downstream of B_D^{place} : for some $m \in \mathcal{M}_D$, metric interpretation first becomes available at B_D^L , and no strictly earlier projection interface $P \prec_D B_D^L$ supports physically interpretable metric use.

The word “first” is therefore not temporal. It means locally minimal in the projection-order dependency relation. This paper does not require global uniqueness of B_D^L ; if several incomparable minimal projection interfaces exist, each is governed by the licensing theorem below. The definite article in informal prose names the interface role; it does not assert global uniqueness beyond the local minimality stated here. The boundary may be represented in a construction sequence, but it is not a dynamical surface, spacetime location, causal event, or measurement interaction.

Definition 5.3 (Generative L-metric reading). A generative reading treats B_D^L as producing, causing, creating, selecting, or ontically grounding interior metric structure.

Definition 5.4 (Licensing reading). A licensing reading treats B_D^L as the minimal legality condition under which already standing-positive content may be described using metric predicates, without changing the role assigned to B_D^{place} or making the metric predicate a source of standing.

Proposition 5.5 (Weak definition leaves room for the theorem). *The definition of a local B_D^L alone does not imply that metric descriptors are interior-only. It states only minimal availability of metric interpretation. The licensing theorem must still exclude boundary-authorizing, generative, transmissive, and retro-authorizing uses.*

Construction witness. The definition quantifies over projection interfaces and metric interpretability. It does not assert non-generativity, no-backflow, or interiority. Those results require the AMetric exclusion theorem, the normal form for boundary authority, and the constrained licensing relation. $\square$

5.1 Anchor, tensor role, and skin at the metric interface

The role discipline classifies interpreted variations. Fix the original governing context $\kappa(c)$ and the complete original-use profile $P(c)$, including partial domains and joint uses. ATS gives the dependent tower of realization, complete content, and context; its fibres need not form independently variable Cartesian factors. A variation changing context has anchor effect; one preserving context but changing a required profile coordinate has tensor effect; one preserving the entire context and profile has skin effect. This is the current ATS, Theorems 3.1 and 4.2 [10]; the older companion is retained as provenance [5].

Definition 5.6 (L-metric anchor). Relative to a chosen local interface B_D^L and the metric uses whose dependency chains pass through it, the L-metric anchor is their original definedness condition under fixed-domain redescription discipline. The prerequisite claim is local to those chains; an incomparable minimal metric interface may support other uses.

Definition 5.7 (Metric tensor role). A metric tensor role, in the broad AASC sense, is complete quotient-level comparison, compatibility, continuation, transport, or gauge-preserving content whose specified variation changes a required original answer or definedness while preserving the governing context. A metric tensor field in differential geometry is a mathematical specialization, not the general meaning of tensor role here.

Definition 5.8 (Metric skin). Metric skin is a variation preserving the governing context and every required original-use answer, including definedness, after lawful input and output transport. A coordinate, unit, diagram, fitted parameter, or raw component is skin only for variations satisfying this test. Preservation of a single standing bit alone does not suffice.

Proposition 5.9 (No role upgrade at the L-metric boundary). *For the same original context and complete use family, a skin variation cannot change a required answer, and a tensor variation cannot change the governing context while still being that same classified incidence. A raw component exhibiting several effects must be refined by its actual variation incidences; it need not have one permanent role name.*

Construction witness. Compare the original context indices first. If they differ, the variation has anchor effect. If they agree, equality or inequality of the complete profiles gives skin or tensor effect respectively. These cases are exhaustive and disjoint for that interpreted variation. A changed required answer contradicts profile equality; a changed context contradicts the tensor case’s fixed context. A symbol or component can participate in several such variations, so exclusive typing of the whole raw component requires a separate homogeneity result. ATS, Theorems 4.2–4.4, constructs the canonical refinement by pairs consisting of the component and its realized effect. This does not classify every numerical parameter as skin or every change of applicability as a change of context. $\square$

6 The L-metric licensing theorem

The weak interface definition leaves the licensing claim to be proved. We construct the relation from original-use semantics and verify all five licensing conditions. No admission is inferred from calculability, and no relation membership is inferred merely from the first coordinate being eligible.

6.1 Original metric-use records and their continuation contract

Let C_D be the set of interpreted candidate metric-use records in the fixed original context. A record c specifies its original act $\alpha(c) \in \text{Eval}_D$, descriptor $\mu(c) \in \mathcal{M}_D$, target interpretation, required result graphs, and the witness and continuation claims it undertakes. The act $\alpha(c)$ indexes the *complete original use*, including its original conditions and required repertoire; changing those promises changes the indexed use. Different faithful witness implementations can represent that same use, but a safe weak promise cannot be projected into a license for an unsafe stronger promise. Its original conditions are fixed before its correctness is tested. The carrier includes unsuccessful proposals; a semantic record need not be a finite certificate or computable data structure. Weak candidates without an established target interpretation remain in $X_D(m)$ without thereby entering an evaluated physical-use claim.

Retain every original question in (2): actual relata and occurrence where required, comparison and output meanings, calibration and unit transport, operation applicability, original joint domains, and historical and continuation obligations. Include the original boundary work: which original qualification is answered, what its target and application conditions are, and which occurrence and joint warrants it preserves. These questions express the functional work recovered by the Kernel, not an assumed label declaring every proposed boundary role admissible or forbidden. Relational content is retained by its full membership graph without selecting an occupant. A reduced interface must establish (3).

Let $A_D(c)$ mean that the original local conditions hold: $Q_D(\alpha(c))$, correctly interpreted and sound result records, the coverage actually required, typing and witness conditions, and availability of operations the task requires. These conditions come from the physical or mathematical task; they

are not defined as membership in Lic_D or as approved mechanism names. A false assertion, wrong calibration, or missing required domain can violate them even when the notation is stable.

Specify independently a relation $R_D \subseteq C_D \times T_D \times C_D$ of declared uses and an original edge-warrant predicate $W_D(c, t, c')$. The latter records actual applicability, target and result interpretation, joint conditions, and retained-history attribution. The repertoire includes every continuation whose preservation is claimed, not only already successful paths. Required unavailability fails readiness in A_D ; it cannot pass by absence of an edge. For every $a \text{ Red}_D a'$, readiness requires its metric-use transport edge, with $\alpha(c') = a'$ and a descriptor $\mu(c') \in \mathcal{M}_D$; the edge warrant includes the independently proved preservation of the original context and complete profile, hence $\mu(c') \sim_D^M \mu(c)$. Thus all redescrptions quantified over in (L2) are required, not a selected successful subfamily. This includes the original boundary-work coordinates, not only the current standing bit. Calling a transformation admissible does not prove that law. Typed output transports preserve identities and composition, so comparison of values does not depend on an undeclared conversion route.

A finite declared path is safe when every node satisfies A_D and every edge satisfies W_D , including the length-zero path. Put

$$S_D = \{c \in C_D : \text{every finite declared path from } c \text{ is safe}\}, \quad (4)$$

$$\Phi_D(U) = \{c \in C_D : A_D(c) \text{ and } \forall (c, t, c') \in R_D, W_D(c, t, c') \text{ and } c' \in U\}. \quad (5)$$

This is the receiving contract for the original continuation claim. An immediate use has no additional physical continuation obligation beyond its stated task, but the required redescription edges and their readiness remain. Eventual success or a physical propagation law needs its own original semantics; local safety does not prove it.

Construction and completeness. The operator Φ_D is monotone and

$$S_D = \Phi_D(S_D) = \bigcup \{U \subseteq C_D : U \subseteq \Phi_D(U)\}. \quad (6)$$

Safety of the zero-length and one-edge paths gives local validity and every edge warrant. Prepending an edge to any successor path shows that a safe record has safe successors. Conversely, local validity and warranted edges to safe successors make every finite path safe. Finally induction along a finite path from any $U \subseteq \Phi_D(U)$ keeps its nodes in U and supplies all warrants, so $U \subseteq S_D$. This proves the equation without finiteness, effective enumeration, or a selected-path hypothesis.

The construction is Structure, Theorem 6.3, and Exhaustion, Theorem 5.3, on the explicit carrier C_D , predicates A_D, W_D , and repertoire R_D [8, 9]. Existence of this set does not prove $S_D \neq \emptyset$. Positive metric realization requires a point of S_D , equivalently a nonempty sound region. A local B_D^L exists only when its own nonempty-interpretation and minimality conditions are established.

Define the canonical licensing relation by

$$\text{Lic}_D = \{(\alpha(c), \mu(c)) : c \in S_D\}. \quad (7)$$

This projects actual safe use records, not arbitrary positive acts. An entry does not validate an incorrect encoding of that entry: an offered presentation or certificate must still realize its claimed original record. A restricted implementation can soundly present a subset; completeness requires coverage of every original licensed use it claims to implement.

Lemma 6.1 (Admissible physical interpretation has qualified original relata). *Let m satisfy the L -metric dependency condition and $x \in X_D(m)$. If its physical interpretation is admissible for the*

original fixed-domain use and continuation repertoire it undertakes, then $\iota_m(x)$ is standing-positive. For ordinary single-domain uses, identifying x with that act gives

$$\text{dom}_D^{\text{phys}}(m) \subseteq \text{Int}_D^+.$$

For a composite the conclusion concerns the independently qualified joint-use act, with all required component and joint-domain conditions retained.

Construction witness. The interpretation supplies the original act and the actual meaning of its qualification condition. Admissibility for this use means those original local and promised-continuation conditions hold, hence supplies a safe record c with $\alpha(c) = \iota_m(x)$ and $\mu(c) = m$. The length-zero condition gives $A_D(c)$ and therefore $Q_D(\alpha(c))$. Equation (1), obtained from original incidence rather than metric licensing, gives $L_D(\alpha(c)) = 1$.

This passage is use-specific. A false metric report can identify its events perfectly while failing its answer-role; its inscription remains determinate. Nonqualification is therefore not identified with an unfixed representative. For a composite, the local condition includes its original joint relation. Separate positive component bits cannot substitute for it. Exhaustion, Theorem 4.8, gives the exact descent test: joint-domain membership must be constant on retained input fibres and defined retained outputs must agree there [9]. The original incidence or a sound constructor theorem must establish the joint condition. $\square$

Theorem 6.2 (L-metric licensing theorem). *Let D have the original-use interpretation and continuation contract above, and the AMetric exclusion discipline of Theorem 4.6. Let B_D^L be a local L-metric boundary and let m become physically interpretable there. Every admissible background same-scope use of m belongs to the constructed Lic_D and satisfies (L1)–(L5). Conversely, a record witnessing (7) supplies an admissible articulation for that original task. The relation contains every sound realization of those same original-use obligations.*

Its role at the interface is licensing of standing-positive interior articulation. The descriptor's licensed use supplies no primitive authority for its own qualification, no primitive selector or self-licensed boundary operation, and no additional boundary channel, retrospective repair, or backward authorization.

Construction witness. An admissible interpreted use has a safe record c , so (7) supplies membership. Verify each required condition.

(L1) *Domain.* Lemma 6.1 gives $L_D(\alpha(c)) = 1$, so $\text{Lic}_D \subseteq \text{Int}_D^+ \times \mathcal{M}_D$.

(L2) *Redescription.* For $\alpha(c) \text{Red}_D a'$, required readiness and the declared original transport supply an edge (c, t, c') . Equation (6) gives its warrant and $c' \in S_D$. The warranted transport gives $\alpha(c') = a'$ and $\mu(c') \sim_D^M \mu(c)$, hence $(a', \mu(c')) \in \text{Lic}_D$. For functional original result coordinates, the two independently sound, complete readers return the same original result after transport, including its definedness. For relational coordinates they preserve the original membership graph. This is original-incidence conservation, not assumed equality of raw components.

(L3) *No boundary mutation.* The warranted redescription edges preserve the original context and complete profile, including the boundary-work coordinates specified above. They therefore preserve which original qualification is answered, its target and applicability conditions, and its occurrence and joint-warrant role. Preservation of that functional work is stronger than equality of one standing bit; equal bits alone would not prove (L3). Its individually warranted qualification reports also coincide by (1). A metric condition already in Q_D retains its original role; asserting its license gives it no extra power to alter that work. A proposed new authority role remains open to examination under the original-incidence and AMetric tests. A justified different question is a

different use, whereas a competing answer to the unchanged question is tested for correctness rather than declared a scope change merely because it differs.

(L4) *Continuation.* The fixed-point equation gives every declared edge warrant and a safe successor. Induction gives compatibility for every finite retained continuation and all joint conditions in the original repertoire. Interior states may change along those edges; replacing the repertoire or its meaning requires a new contract and proved transfer. Neither present qualification alone nor a finite sample of successful paths establishes this universal result.

(L5) *Non-generativity.* The original condition $Q_D(\alpha(c))$ is a ground of $A_D(c)$ before the licensing relation is constructed. The relation records qualified metric articulation; it does not produce the original incidence, confer truth on its condition, or supply a missing application premise. The AMetric fibre and grounding arguments exclude using its own asserted entry as new qualification authority.

Conversely, a witness $c \in S_D$ meets every original local and path obligation, so supplies a sound metric-use realization. Every sound region $U \subseteq \Phi_D(U)$ lies in S_D ; its relation entries already belong to Lic_D . This establishes completeness for the independently specified task. The selector, operation, transfer, and historical exclusions use the original-incidence, fibre, grounding, and retained-occurrence arguments, made explicit for backflow in the next section. No step introduces an intervention input, message mechanism, or new dynamics at the interface. $\square$

Corollary 6.3 (Boundary non-generativity). *The L-metric boundary licenses metric description; it does not generate metric structure, standing, or physical ontology.*

Construction witness. The construction takes original-use incidences, descriptor meanings, and a continuation contract as grounds and returns their licensing relation. Its result is an eligibility judgment, not an operator producing a metric or physical bearer. If the original conditions determine a unique metric, that determination belongs to those conditions and their proof, not to the bare act of licensing. If several metric realizations satisfy them, the relation retains all warranted realizations and selects none. A constructor or physical production process can be added, but its application has its own original domain and warrant. Treating its asserted license as its ground is the self-authorization excluded by Theorem 4.6. Constitutive non-generation does not prohibit new interior constructions or later successful trials. $\square$

Corollary 6.4 (No metric transmissivity). *The licensing relation introduces no channel by which standing, information, metric structure, or causal influence is transmitted between physical loci. An interior channel requires its own admitted input, intervention, transport, and output incidences.*

Construction witness. Equation (7) supplies qualified description records and their originally warranted uses, with no additional intervention or channel law. A channel already specified in R_D can be licensed on its actual conditions; relation membership alone does not supply its warrant. Recovering an extra target answer from unchanged source data must pass the fibre test; transferring qualification must establish the target incidence. The conclusion concerns what the boundary supplies, not the impossibility of interior communication. $\square$

Remark 6.5 (Status and existence of the theorem). Kernel necessity applies to the actual incidences whether or not their proposed metric uses qualify. The theorem constructs the canonical licensing relation for the interpreted task. Positive realization and existence of a minimal interface are separate questions. The proof applies at any admitted metric projection interface; local minimality identifies the first such interface in dependency order when one exists. It proves no unique global interface, unique metric, or unique physical occupant.

Remark 6.6 (Scope of the licensing theorem). The theorem does not derive field equations, Hilbert-space no-signaling, the Standard Model parameter table, a cosmological-constant value, or cosmological model selection. Those require their original physical conditions and source-to-target warrants. It also does not assume that a present qualification bit retains every future or joint-use distinction.

Conservative receipt of an established original use. The construction also receives an independently proved consumer use without adding a physical premise. Let (Y, A_Y, R_Y, W_Y) retain that consumer's complete original use records, local conditions, declared operations and edge warrants. Suppose its source proof establishes a region $V \subseteq Y$ whose records satisfy A_Y and whose every declared outgoing edge is warranted and ends in V . Let h interpret those records in C_D as the same original uses. This interpretation retains every required question, result graph, original incidence, joint domain and promised operation, including the original guards and all required redescription availability. It may retain source information in addition to the metric readout: $\mu(h(y))$ need not determine the complete joint source by itself.

Assume the receiving interpretation proves local validity transfer $A_Y(y) \Rightarrow A_D(h(y))$ on V . For every target edge $(h(y), t, c') \in R_D$, require an original source edge $(y, s, y') \in R_Y$ with $c' = h(y')$ and a proved transfer of its warrant to $W_D(h(y), t, c')$. Every originally promised consumer operation must also be represented; omitting one would change the use. Then

$$h[V] \subseteq S_D, \quad \{(\alpha(h(y)), \mu(h(y))) : y \in V\} \subseteq \text{Lic}_D.$$

Thus an actual point of V supplies a positive licensing witness. Neither injectivity of the metric readout nor coverage of unrelated target records is required.

Construction witness. Put $U = h[V]$. For $c = h(y)$ with $y \in V$, the independently established source conditions and local transfer give $A_D(c)$. Any declared target edge from c lifts to an original source edge. Source closure supplies its warrant and $y' \in V$; warrant transfer gives $W_D(c, t, c')$ and $c' = h(y') \in U$. Consequently $U \subseteq \Phi_D(U)$. Equation (6) gives $U \subseteq S_D$, and equation (7) gives the displayed licensing entries. This is the one-way sound-region transport after Structure, Theorem 6.8 [8], applied to the original metric-use records.

When the receiving records simply retain the original consumer records and interpret their actual operations, the lift is that same operation and its warrant is the original proof. This observation does not prove any consumer's physical premises: its original-use interpretation, required availability and closure must be established at the source. It also does not authorize replacing an independently fixed receiving repertoire by a smaller one. An immediate original task gains no further physical future from this receipt; a task that promises continued validity retains every such obligation. A legal step type need not preserve standing, and a guarded theorem about coherent trajectories does not become a theorem about all legal futures. Nonempty realization and local interface minimality remain separate, as above. $\square$

6.2 Exact transport and joint-use checks

Aggregate local validity and outgoing edge warrants into B_X, B_Y , let $\text{Succ}_X, \text{Succ}_Y$ be the declared successors, and let $f : X \rightarrow Y$ have an independent original interpretation. Exact transport of the universal-safety operators holds precisely when

$$x \in B_X \iff f(x) \in B_Y, \tag{8}$$

$$x \in B_X \implies f[\text{Succ}_X(x)] = \text{Succ}_Y(f(x)). \tag{9}$$

Substitution gives $\Phi_X(f^{-1}V) = f^{-1}(\Phi_Y(V))$ for every $V \subseteq Y$. Conversely, apply this equation to $V = Y$ and then to each successor set to recover the displayed conditions. Matching finite paths gives $S_X = f^{-1}(S_Y)$. Moreover $f[S_X] = S_Y \cap f[X]$, so coverage of the whole target interior requires exactly $S_Y \subseteq f[X]$; surjectivity of $f : X \rightarrow Y$ is sufficient but not necessary. These are Unus, Theorem 5.3 [11]. Forward matching alone can miss an unsafe new target continuation. Label-sensitive observations and metric readouts require their own preservation equations as well.

Separate safety for two repertoires likewise need not give safety for interleavings. The joint task uses the intersection of local validity conditions and the union of successors. Its greatest safe region is contained in the intersection of the separate regions, with equality exactly when that intersection is closed under every declared successor (Unus, Theorems 6.1–6.2). A joint metric use or cross-fixation comparison therefore cannot inherit a license merely because its ingredients separately qualify.

7 No metric backflow

The licensing theorem does not prevent new evidence, sound constructors, or successful new trials. It prevents an interior description from supplying its own missing qualification merely by being used. The original occurrence and complete use question determine which non-repair claim is at issue.

Definition 7.1 (Metric backflow). Metric backflow is the purported creation or alteration of qualification at an unchanged original locus by an interior metric result whose asserted authority supplies no independent warrant for that locus. It includes replacing a failed original occurrence by a later successful one while retaining the original attribution, or claiming an extra source-level distinction from data that do not retain it.

7.1 Admissibility standing and empirical confirmation

Evidence can improve knowledge of an original qualification and support a new physical or inferential use. Such a use has actual target-specific conditions. A sound model, measurement, or constructor theorem can establish them without a separate chronological checking event. What evidence cannot do is make an originally false qualification true while the original bearer, occurrence, and complete conditions remain unchanged. In particular,

later discovery of an original qualification $\neq$ replacement of its original conditions.

A distinction between empirical warrant and standing therefore does not dismiss scientific evidence. It identifies the original incidence and transfer theorem that make an evidential use warranted.

Theorem 7.2 (No-backflow theorem). *An interior metric fact cannot generate or alter qualification at an unchanged original incidence, or supply its own missing target-specific license by circular dependence. A claimed backward determination from retained data exists exactly under the AMetric fibre criterion; when it exists, its original warrant remains necessary. Later evidence and independently sound constructions can establish previously unknown qualification or a qualified new use without violating this result.*

Construction witness. Fix an original act a , including its occurrence and complete question. Every faithful report of its qualification records the same $Q_D(a)$ by (1). Replacing the original data yields a different act, whereas retaining them retains that truth value. For a retained history $(a_0, \dots, a_n)$, a failed original conjunct in $\bigwedge_i Q_D(a_i)$ remains failed when further acts are appended. No assumption that all future endpoints must remain negative is used.

Next let $p : Z \rightarrow X$ be the actual forgetting map from the richer metric data and $w : Z \rightarrow V$ the target answer claimed to be supplied by the unchanged source. If a fibre has distinct required answers, no source-only decoder exists. If all fibres agree, the unique descended answer adds no distinction absent from the retained data, and its semantic warrant must still be shown. A metric, numerical score, or probability supplies no exemption from this criterion.

Finally, if the offered authorization consists only of an unseeded family C in which every rule concluding a member already requires a member, the complement argument in Section 4 excludes that family from grounded closure. A warranted external entrance defeats the obstruction's premise and may legitimately establish the use. These are the separate historical, information, and support arguments of Kernel, Lemma 3.3; AMetric, Theorems 7.2–7.5; and Structure, Theorem 7.2 [6, 7, 8]. They do not infer negative qualification from an unsuccessful search or an incomplete proof. $\square$

Corollary 7.3 (Ontology-from-fit collapse). *Empirical success can support an independently warranted physical interpretation, model retention, or refinement. Fit alone does not make the model's metric objects a primitive source of the qualification required for that fit to answer its target question.*

Corollary 7.4 (Geometry-from-success collapse). *Covariance, geometric elegance, predictive compression, or formal unification does not by itself establish a missing original-use incidence. A proof that such a property entails the required qualification is a legitimate domain theorem and must retain its actual conditions.*

Corollary 7.5 (No horizon backflow). *A horizon, singular surface, cutoff, or causal boundary in an interior model does not by its name or geometric status become the AMetric or L-metric licensing interface. Any claimed identification must establish the same original question, conditions, and preservation role.*

Remark 7.6 (Scientific warrant and new endpoints). A repaired apparatus can pass a new trial, and a later check can discover that an unchanged proof was valid all along. Neither changes an earlier occurrence's original qualification. Preservation of an all-stages history is distinct from negative absorption of endpoint states; the latter requires its own transition condition (Exhaustion, Theorem 6.2) [9].

8 Locality as licensed interior structure

The realized-physics arc distinguishes bounded standing-localization and continuation neighborhoods from later metric distance, field support, and spacetime regions [13, 14]. Pre-metric role locality can be physically significant. Its original realization and its use as an authority over the qualification of that very realization must not be conflated.

Definition 8.1 (Licensed locality). Locality is licensed when its dependence, perturbation, restoration, and comparison claims satisfy the original conditions of the admitted physical use. In the realized-physics arc these are expressed through standing-localization classes and local continuation domains before a numerical distance or propagation law is supplied.

Definition 8.2 (Locality role). A locality role is a relation, neighborhood, support, dependence, or transport-path condition constraining the retention and comparison of physical content. It need not be metric distance, a manifold topology, a factorized state space, or a finite-speed law.

Theorem 8.3 (Locality licensing theorem). *Metric locality requires admitted relata, comparison structure, and its target-specific preservation conditions. Pre-metric standing-localization can precede*

that metric articulation under the hypotheses of the realized-physics arc. Neither form of locality supplies primitive self-authorization for the original physical use whose conditions it presupposes.

Construction witness. A metric locality claim is an original relation among its identified loci or comparisons. Its admissible use therefore has a record in the construction of Section 6; its existence and readouts are not supplied by the licensing relation itself. The source's earlier bounded-localization result is conditional on realized persistence and its restoration/non-collapse conditions (R2, Theorem 7.1). The continuation-neighborhood result additionally retains compatible continuation and boundary fixation (R3, Theorem 7.1). Those original hypotheses supply their respective pre-metric roles; no absence of a metric is used to deny those roles. Finally, a locality relation offered as the source of its own missing license is subject to the original-incidence, fibre, and grounding tests of Theorem 4.6. An independently warranted local condition can form part of the original physical task without becoming an extra self-authorizing boundary power. $\square$

Corollary 8.4 (Locality and pre-metric joint standing). *A pre-metric joint-standing relation is not refuted merely by failure of a later metric-locality condition whose application has not been established. Any locality constraint actually belonging to the original joint-use task must nevertheless be satisfied.*

Remark 8.5 (Bell locality and operational locality). Bell-type assumptions concern specified variables, interventions, and outcome records inside an admitted probabilistic model. Their mathematical consequences require those original assumptions. The licensing result neither removes them nor derives them from pre-metric joint standing.

Corollary 8.6 (Bounded transport and finite propagation). *The arc's pre-metric bounded transport constrains perturbation to admitted continuation-neighborhood paths. A quantitative finite-speed law is a later interior result requiring its own temporal, metric, and dynamical conditions; it is not supplied by the bare licensing boundary.*

R5, Theorem 9.1 and Section 10, distinguishes path/witness-bounded transport from a finite velocity or a supplied causal dynamics [16]. The distinction prevents a structural transport premise from silently acquiring a stronger metric conclusion.

Proposition 8.7 (Exact locality-extension criterion). *Fix an original metric-use contract with operator Φ and greatest safe region S . Suppose a proposed extension retains that interpretation and adds independently specified local obligations B_j^+ and successors Succ_j^+ . Put*

$$\Psi_j(U) = \{x \in B_j^+ : \text{Succ}_j^+(x) \subseteq U\}, \quad \Phi^+(U) = \Phi(U) \cap \bigcap_j \Psi_j(U).$$

If S^+ is its greatest safe region, then $S^+ \subseteq S$, with equality exactly when

$$S \subseteq B_j^+ \quad \text{and} \quad \text{Succ}_j^+(x) \subseteq S \quad \text{for every } j \text{ and } x \in S.$$

The added obligations include their original edge warrants and any required readiness.

Construction witness. Every Φ^+ -sound region is Φ -sound, giving $S^+ \subseteq S$. If the displayed conditions hold, S is sound for every added obligation and hence for Φ^+ , so greatestness gives $S \subseteq S^+$. Conversely, equality and the fixed-point equation for S^+ imply each displayed condition. This is Unus, Theorem 6.4, on the stated receiving contract [11]. $\square$

This exact criterion does not eliminate hidden sectors, new interactions, or new degrees of freedom merely by naming them. A claim that an extension is redundant requires equality of the complete original-use profile; an exclusion based on an exhausted Standard Model bookkeeping space additionally requires that exact space and its maximality theorem. Global compatibility or nonfactorizing tensor content is not itself a locality violation; R2, Sections 14.3–14.5, explicitly retains such possibilities. A changed original task needs its own admission or a justified transfer rather than this unchanged-contract equation.

Part II

Physics-bound audits

9 Fixation-local metric articulation and Lorentzian examples

An admissible physical interval comparison requires the original identification of its relata and the conditions of the particular comparison. Candidate formulas remain weakly interpretable before that physical application is established. Cross-fixation comparison requires a warranted interpretation connecting the original uses; the existence of a metric in one model does not supply that connection.

Definition 9.1 (Fixation-local metric articulation). A metric articulation is fixation-local when its lawful use is confined to relations whose relata, standards, and comparison rules are already fixed within one standing-positive domain.

Definition 9.2 (Cross-fixation comparison). A cross-fixation comparison compares, transports, identifies, synchronizes, or aligns metric structure across distinct original fixations. Its target qualification and comparison conditions must be supplied; the definition does not prejudge whether they are satisfied.

Theorem 9.3 (Fixation-local metricity). *If a fixation admits metric articulation, the license applies only to relations whose evaluation is internal to that fixation. Cross-fixation comparability, transport, or identification is not licensed by metricity alone.*

Construction witness. A cross-fixation metric statement concerns a target use different from mere metric availability in the source. Let h be its proposed original-use interpretation. Source qualification transfers only if the original conditions prove $Q_D(a) \Rightarrow Q_F(h(a))$; equality of positive bits supplies no such implication. A matching rule can establish it when its domain, eligible outputs, and original incidence are warranted. For robust continuation, the target-successor and readout conditions of Section 6 must also be established. Thus a lawful transfer is allowed, while source metricity alone leaves the required connection unproved. This is the precise target-use export condition of AMetric, Section 7.2 [7]. $\square$

Corollary 9.4 (No global metric smuggling). *Global spacetime, global simultaneity, universal metric comparability, global chart structure, or cross-domain identification cannot be inferred merely from local metric articulation.*

9.1 Minkowski interval

Example 9.5 (Lorentzian interval as interior articulation). Let two record events e_1, e_2 have their admitted original interpretations in a supplied Lorentzian model, with their joint comparison

conditions established. In a supplied inertial chart with one spatial coordinate, their interval is represented by

$$s^2 = c^2 \Delta t^2 - \Delta x^2$$

in a chosen inertial chart, or more generally by a Lorentzian model (M, g_{ab}) with the records represented as points or localized regions. Under the L-metric reading, neither M , g_{ab} , Δt , Δx , nor s^2 supplies the boundary standing of the records. The interval is a licensed comparison descriptor applied to already fixed relata.

Proposition 9.6 (Interval interiority). *A metric interval assigned between standing-positive records is an interior comparison descriptor, not a pre-boundary authorizer of the records it compares.*

Construction witness. The interval's original use includes its identified relata, supplied metric and comparison convention, and domain of physical interpretation. A warranted comparison satisfies those conditions and therefore has a licensing record by Theorem 6.2. The numerical interval does not by itself supply its own missing target interpretation or joint-use warrant. A geometric condition can legitimately enter the original physical qualification, and a sound calculation can establish it; neither is primitive self-authorization. Standing of each record separately is not a proof of the full joint comparison conditions. $\square$

9.2 Lorentzian model role classification

For a Lorentzian model (M, g_{ab}) , the L-metric audit is not that the model is unreal. It is that each object must be assigned the correct role.

Object	L-metric/AASC role
M	Carrier or presentation manifold for licensed physical events; not by itself a source of boundary standing.
g_{ab}	Metric comparison and causal-cone structure inside the licensed interior.
Events p, q	Representatives of standing-positive records or localized report classes, not primitive pre-fixational atoms.
s^2 or proper time	Interior invariant comparison descriptor over already fixed relata.
Coordinate chart	Skin unless tied to observable calibration or continuation constraints.
Causal cone	Interior causal/transport descriptor after L-metric licensing.
Geodesic distance or extremal proper time	Interior comparison/transport descriptor, not boundary authority.
Einstein equation	Interior dynamical constraint relating metric and stress-energy representations; not the boundary source of standing for records.

The excluded moves are unsupported ones: asserting that g_{ab} supplies its own missing record interpretation, inferring global cross-fixation comparison without the corresponding map and warrants,

or identifying geodesic completeness with a different qualification question without a proof. A physical model can predict new records or test a geometric condition through an independently warranted inference. The licensing theorem does not prohibit that scientific use.

Remark 9.7 (Why the examples are intentionally modest). The examples do not derive special relativity or general relativity. They show how familiar metric expressions are located by the licensing theorem. The interval and Lorentzian metric may be central to physical modeling; they are not the boundary source of the records they relate.

10 Zero-point vacuum energy as a metric-backflow audit

The zero-point vacuum-energy arc gives a quotient-descent audit. Target instantiation: a purported vacuum-energy prediction is reused across renormalization presentations as the same physical quantity. The audit shows how a standard QFT quantity may be computationally meaningful while failing to descend to the quotient that defines standalone predictive content. The exhaustion question is whether an absolute homogeneous offset survives quotient-preserving redescription as standalone predictive content.

The target is narrow. The section concerns scalar quantities constructed from the homogeneous metric-proportional component of the renormalized vacuum stress tensor in a fixed-background class. It does not classify the full stress tensor in arbitrary curved backgrounds, anomalies, state-dependent curvature terms, material Casimir modeling, or the naturalness problem for the measured effective cosmological constant. In curved-spacetime QFT, renormalization also involves local conserved curvature terms, possible scale dependence, and convention-dependent sign choices. The point here is narrower: the absolute homogeneous matter-sector offset alone does not descend to a standalone prediction without quotient-stable matching data.

10.1 Quotient-descent criterion

Definition 10.1 (Witness-preserving redescription quotient). Let S be a fixed physical sector and let $\sim_S$ identify descriptions related by admissible scheme, coordinate, gauge, normalization, and unit transformations that preserve the empirical witness ledger and boundary status. Output comparison is fixed independently of the proposed prediction. For a redescription $r : D \rightarrow D'$, let $\tau_r : Y_D \rightarrow Y_{D'}$ be the admitted transport of output values, with $\tau_{\text{id}} = \text{id}$ and $\tau_{s \circ r} = \tau_s \circ \tau_r$ wherever composition is defined. A quantity assignment $X(D) \in Y_D$ is compatible with the quotient when

$$X(D') = \tau_r(X(D)).$$

For an invariant scalar output whose physical meaning and units are held fixed, the output transport is the identity and this becomes ordinary quotient descent:

$$D \sim_S D' \implies X(D) = X(D').$$

For dimensionful quantities, the comparison instead includes the independently specified conversion of units. Covariant components likewise require the admitted component transport. The maps τ_r may not be chosen retrospectively to make an arbitrary assignment pass the test.

Theorem 10.2 (Quotient-descent criterion for physical prediction). *A metric-interior scalar X , offered as an invariant physical output with fixed meaning and units, is eligible as standalone predictive content only if it descends to the standing-preserving quotient of the fixed physical sector. More*

generally, the physical output must obey the declared output-transport rule above. If the output changes after that transport under witness-preserving same-scope redescription, then the raw assignment is a presentation coordinate rather than a standing-bearing standalone prediction. It may still function as a scheme-relative calculational quantity, a relational observable, a matched effective parameter, or a boundary-conditioned comparison input. Descent is necessary; it does not by itself determine the value or prove that the sector predicts it without empirical matching.

Construction witness. A standalone prediction must assign the same physical output to presentations of the same target. With output comparison fixed, this means $X(rD) = \tau_r X(D)$: otherwise one target receives incompatible outputs solely because its presentation changed. For a scalar with trivial output transport, define $\bar{X}([D]) = X(D)$. This definition is independent of the chosen representative exactly when X is constant on each $\sim_S$ -class, proving the descent criterion. For nontrivial output transport the corresponding condition identifies $(D, X(D))$ with $(rD, \tau_r X(D))$; a change of numerical units is therefore not a failure of physical invariance. This construction removes representative dependence but supplies neither a numerical value nor a derivation of that value. Relational and matched quantities must still satisfy their own witness and matching conditions. $\square$

This criterion instantiates the presentation discipline required by L-metric licensing. Its receiving data—physical sector, witness ledger, redescription class, and output transport—come from the physical model. The ZPE companion states scalar invariance after transport of units and presentation conventions [30, Definition 7]; the physics-native source also separates invariance from determination without matching [29, Definition 2]. Neither that model data nor a predictive value is generated by the licensing theorem.

10.2 Homogeneous zero-point sector

For one bosonic degree of freedom, a cutoff-regulated zero-point expression is schematically

$$\rho_{\text{zp}}(\Lambda_{\text{UV}}) = \frac{1}{2} \int^{\Lambda_{\text{UV}}} \frac{d^3 k}{(2\pi)^3} \sqrt{k^2 + m^2}.$$

The controversial inference is not the appearance of this regulated expression. It is the step in which its absolute value is treated as a standalone cosmological prediction.

In the restricted homogeneous sector, write the metric-proportional component of the renormalized vacuum stress tensor as

$$\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}.$$

With units, background class, non-vacuum normalization conditions, and the remaining counterterm conventions fixed, finite renormalization permits the same-scope redescription

$$\rho_R \mapsto \rho_R + \delta\rho.$$

In semiclassical gravity,

$$G_{\mu\nu} + \Lambda_R g_{\mu\nu} = 8\pi G_R \langle T_{\mu\nu} \rangle_R,$$

so the simultaneous transformation

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho$$

leaves the total coefficient

$$\Lambda_{\text{eff}} = \Lambda_R + 8\pi G_R \rho_R$$

unchanged, with sign convention fixed by the field equation above. The gravitational witness is preserved by the simultaneous change of the pair (Λ_R, ρ_R) , not by changing ρ_R while holding the gravitational split fixed. This is the receiving match to the restricted homogeneous sector in the three ZPE sources [29, Theorem 1 and Section 5][30, Proposition 2 and Theorem 3][31, Section 6.1].

The tested output is an absolute physical scalar in fixed units, so its output transport under this change of subtraction convention is the identity. The renormalized coordinate ρ_R itself may instead be carried equivariantly by $\rho_R \mapsto \rho_R + \delta\rho$ for calculation. That valid coordinate transport does not turn the coordinate into an invariant absolute prediction.

Theorem 10.3 (Zero-point vacuum energy is not a boundary-standing generator). *Within the restricted homogeneous metric-proportional vacuum sector V_0 , no nonconstant scalar depending only on the absolute matter-sector coefficient ρ_R descends to the finite-renormalization quotient*

$$\rho_R \sim \rho_R + \delta\rho.$$

Therefore ρ_R , treated as an absolute single-interior offset, is not standalone predictive content and cannot function as a boundary-standing generator. The compensated shift is a skin variation for this sector's complete physical-use profile; the split coordinate is not thereby declared devoid of every calculational role. An admitted witness fixing the offset would change the stipulated quotient.

Construction witness. The translation orbit of any value ρ_R under $\rho_R \mapsto \rho_R + \delta\rho$ is all of $\mathbb{R}$. Any scalar $X(\rho_R)$ invariant under all such shifts is constant on that orbit. A constant on the quotient carries no information about the absolute value of ρ_R . Hence a nonconstant absolute zero-point coefficient does not descend to the quotient and cannot be a standalone prediction. In semiclassical gravity, the compensating shift of Λ_R leaves Λ_{eff} fixed, so curvature observations determine the total effective coefficient rather than the split (Λ_R, ρ_R) . A new witness fixing the split would restrict the admitted shift equivalence and require a fresh receiving-sector check; it would not contradict the result on the original quotient. The failed step is not calculation of a regulated contribution; it is treating a non-invariant interior coordinate as boundary-stable predictive content. $\square$

Corollary 10.4 (Zero-point comparison as failed prediction-forming procedure). *The direct comparison*

$$\rho_{\text{zp}}(\Lambda_{\text{UV}}) \quad \text{versus} \quad \rho_{\Lambda, \text{obs}}$$

is not a failed standalone prediction of QFT. It is a failed prediction-forming procedure: a regulator- and subtraction-dependent coordinate is compared with an empirical gravitational parameter as if both occupied the same quotient-stable role.

10.3 Role table

ZPE/QFT object	L-metric/AASC role
Regularization scheme	Presentation choice; skin unless tied to a fixed observable.
Subtraction convention	Skin; admissible redescription in the restricted sector.
Absolute ρ_R	Non-invariant ledger coordinate for the homogeneous matter-sector offset.

ZPE/QFT object	L-metric/AASC role
$\rho_R \mapsto \rho_R + \delta\rho$	Matter-sector shift in V_0 ; gravitational witness preservation requires the simultaneous compensating shift below.
$\Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho$	Compensating gravitational parameter re-description preserving the effective coefficient.
Λ_{eff}	Matched effective gravitational coefficient; curvature witnesses determine the total, not the split.
Casimir force	Relational response witness; not a measurement of an absolute vacuum offset.
Phase-transition energy difference	Relational standing content between dynamically connected phases.
Curvature observation	Witness of the total effective coefficient, not of ρ_R alone.
Naturalness/radiative stability	Surviving boundary-input and matching problem for explaining the small measured Λ_{eff} .
Naive zero-point comparison	Failed prediction-forming procedure; metric backflow if treated as boundary authority.

10.4 Surviving physics

Casimir forces, phase differences, transition actions, response functions, and curvature-dependent quantum effects are not removed by the theorem. They survive when their witness structure is relational or when the relevant parameter is fixed by an admitted matching condition. The measured Λ_{eff} remains a real empirical target and a naturalness/matching problem for effective field theory and gravity [47, 50, 56]. What is removed is only the claim that an absolute cutoff zero-point sum is already a standalone cosmological prediction. Failure of quotient descent does not make a scheme-relative quantity useless; it blocks only standalone prediction status unless additional witness-fixing or matching data are supplied.

Remark 10.5 (Why this belongs in the L-metric manuscript). The zero-point case is a concrete no-backflow and exhaustion audit. An interior calculational coordinate is pushed into prediction status without descending to the appropriate quotient. The L-metric theorem classifies the error: metric-interior usefulness is being promoted into boundary-stable standing. The quotient-descent criterion prevents that promotion while preserving the relational and matched physics that actually survives.

11 Physics-bound audits of licensed interior description

Each audit fixes a physical use and examines its warrants: the original relata, admissible changes of presentation, required comparisons, and promised continuations. The question is whether the proposed use satisfies those requirements. These applications illustrate the licensing theorem; they supply neither its premises nor cumulative evidence for AASC.

Only some audits are directly metric-descriptor cases. The Lorentzian, ZPE, Hubble, and constants cases concern metric or metric-dependent descriptors directly. The Standard Model audit is included as a structural-licensing case: it illustrates how a licensed interior physical architecture can have structural standing while numerical parameter completion remains downstream.

11.1 Structural/parametric separation: the Standard Model audit

The Standard Model example separates an admitted physical architecture from the numerical values within it. Gauge-product form, representations, and operator availability specify structural content; couplings, masses, and mixing coordinates require their own warrants. The frozen companion makes this distinction in Criterion 1.6, Definition 1.7, and Theorem 12.2 [32]. This audit uses that distinction without importing the companion’s full structural derivation. Appendix D records its tuple and source-specific restrictions.

Lemma 11.1 (Structural/parametric separation). *A licensed physical structure may be standing-positive at the structural level while its metric or numerical parameter table remains downstream. L-metric licensing admits the standing-preserving structural interior; it does not automatically promote every numerical coordinate, fitted parameter, scale value, or coupling value inside that interior into a primitive prediction.*

Construction witness. A structural datum changes the admissible object class, role ledger, witness family, operator-availability ledger, transport ledger, anomaly/vacuum survivor ledger, or boundary bookkeeping. A parameter datum changes a value inside that fixed structural occupant without changing those ledgers. If a numerical value is promoted merely because the structural occupant is licensed, then the promotion treats interior metric display as standing authority. That is exactly metric backflow. Therefore structural standing and parameter closure must be typed separately. $\square$

Corollary 11.2 (Numerical humility from structural licensing). *The derivation or admission of a structural physical architecture does not by itself derive its full numerical parameter table. Numerical values require their own quotient descent, witness preservation, exactness certificate, calibration-removal theorem, or scale-anchor derivation.*

Licensing can therefore apply to an entire physical architecture while leaving its numerical parameters to separate determining arguments.

11.2 Cosmological transport closure and the Hubble comparison

The Hubble example asks when early- and late-regime inferences concern the same physical parameter. The comparison requires an explicit transport of the target, observations, calibration records, inference maps, and errors. Numerical agreement alone does not establish that identification. The companion analysis supplies a concrete application of this test [33].

In that analysis, the early-sector parameter block supplied by the Planck/CAMB calculation is treated as an anchor. The late-sector quantity is represented by

$$H_0^{AASC} = H_0 + \Delta T(0),$$

where $\Delta T(0)$ is the present-epoch transport correction in the implemented basis. Its penalty measures departures from the declared comparison conditions. Those conditions, stated below, must include every input that changes the comparison’s required output.

Lemma 11.3 (Metric transport-closure). *A metric parameter inferred in one observational regime may be compared to a metric parameter in another regime as the same physical quantity only if the transport map preserves anchor identity, tensor witnesses, quotient observables, continuation structure, calibration records, declared estimator roles and error discipline, and undeclared-hidden-remainder closure. Preservation is up to the independently specified lawful equivalences of the inference tuple. Otherwise the comparison is an unclosed transport identification rather than a completed same-parameter metric claim.*

Construction witness. A cross-regime parameter comparison is determined by its target, witness content, observable extraction, transport, calibration records, inference maps, and error discipline. Holding their physical roles fixed under the declared transports fixes what is being compared and how agreement or discrepancy is evaluated. Changing a role without its transport changes that comparison. Estimator preservation does not require the two observational pipelines to use identical algorithms, nor does target identity require their finite-data estimates to coincide. It requires the declared inference maps and their transported statistical and systematic errors to remain the maps for this comparison. Any additional direction that changes its standing-bearing output must also be included in the relevant ledger. Hence a desired late-sector value alone cannot establish the same-parameter identification. This is the receiving match to the external inference tuple and transported error comparison in [26, Definitions 3.1–3.5 and 4.2], and to the numerical tuple in [33, Definitions 5.1 and 6.1]. $\square$

The frozen companion reports a constrained fit under its chosen transport basis, penalty, data stack, and native Planck validation likelihood; it does not use the full `click/clipy` production likelihood. SH0ES calibration information enters the joint likelihood, so the SH0ES comparison is a residual within that fit, not independent validation. The source’s Section 15 contains the numerical results and diagnostics [33, 34, 38, 35, 36, 37]. Its reported non-resolution is conditional on that implementation. It neither rejects SH0ES nor adjudicates unrestricted Λ CDM or all new-physics extensions; no unrestricted baseline was retained for that comparison. Finite diagnostics prove neither exact zero residual nor semantic completeness beyond the tested task.

Corollary 11.4 (No metric repair by cross-regime numerical agreement). *A late-time metric calibration cannot function as boundary repair for an early-universe anchor merely by producing a desired numerical value. If the transport ledgers do not close, the result may be a useful diagnostic or a prompt for new model structure, but it is not a completed same-scope metric identification.*

11.3 Synthesis of the physics audits

Audit case	L-metric classification lesson
Lorentzian interval	Admissible comparison requires the original relatum and joint-use conditions; the reported interval cannot supply its own warrant.
Zero-point vacuum energy	Failure of the required output transport blocks the claimed standalone prediction; relational and matched quantities retain their own witness conditions.

Audit case	L-metric classification lesson
Standard Model structure	Structural standing is distinct from numerical parameter closure.
Hubble transport closure	Comparing cross-regime values as the same physical quantity requires justified target, calibration, inference, and error transport.
Entanglement/no-channel	Joint standing does not supply controllable metric signaling.

These audits apply the semantic tests of Sections 3 and 6. Fix the original question and its required uses, then test the proposed answers, retained distinctions, supporting warrants, and promised continuations. A failure needs a specific witness: an incorrect or omitted required result, a lost distinction, invalid offered support, or an unsafe promised use. Mechanism names are not failure proofs; a lawful selector, operator, or constructor remains admissible on its actual conditions. A changed question requires a separately interpreted task, while a missing warrant leaves a claim unestablished rather than proving it physically false. The audits illustrate these obligations without becoming premises of the licensing theorem.

12 Entanglement, joint standing, and no boundary-supplied channel

Target instantiation: outcome records and joint distributions are reused as standing-bearing physical records across measurement descriptions. The L-metric boundary classifies Bell-type correlation without adding a collapse mechanism or deriving Hilbert-space no-signaling: joint compatibility can be fixed before metric-local outcome records are articulated, but the boundary supplies no controllable communication channel [24, 25].

The source/receiving match is specific. The entanglement source fixes a bipartite preparation, setting and outcome interfaces, the operational table $\mathbb{P}(a, b \mid x, y)$, and its no-signaling marginals as the Bell witness object (Section 2.1). Its local realization uses declared witness channels (Proposition 4.3). The record-construction source supplies the separate requirement that a record's role be fixed for same-domain reuse; its full measurement-interpretation classification is not imported here. The present application retains those witness identities. It does not infer a probability law, a family of physical measurements, or no-signaling marginals from the mere existence of a standing class.

Definition 12.1 (Joint AMetric standing). A composite has joint AMetric standing when the evaluated object is admitted as a single standing-positive composite before metric separation among later records is articulated.

Definition 12.2 (Metric-local outcome record). A metric-local outcome record is a record represented inside the licensed metric interior as located at a local domain, detector channel, spacetime region, or measurement interface.

Proposition 12.3 (Joint standing projection). *When an admitted physical model supplies record projections of a jointly standing AMetric composite into separated metric-local records, L-metric licensing does not require an additional boundary transmission between those records to authorize their joint standing.*

Construction witness. Fix the composite, its admitted record projections, and the common witness identity. The projections present the joint content through the supplied local interfaces; their

separation does not supply a new qualification of that content. A proposed boundary transmission would add an action or transfer to the qualification interface, which L-metric licensing does not provide. The joint standing therefore requires no such additional boundary transmission. The model still bears the existence and dynamical burdens for its record projections and any interior processes relating them. $\square$

12.1 Correlation, intervention, and boundary classification

Let x, y be local setting inputs, a, b local outcome records, and $\mathbb{P}(a, b \mid x, y)$ an interior probability descriptor inside an admitted experimental regime. Three claims must be kept distinct.

(i) *Correlation claim.* The joint distribution may fail to factorize:

$$\mathbb{P}(a, b \mid x, y) \neq \mathbb{P}(a \mid x)\mathbb{P}(b \mid y).$$

(ii) *No-channel claim.* Receiver marginals remain independent of the remote controllable input:

$$\mathbb{P}(b \mid x, y) = \mathbb{P}(b \mid y), \quad \mathbb{P}(a \mid x, y) = \mathbb{P}(a \mid x).$$

(iii) *Boundary classification claim.* Nonfactorizing correlation is compatible with joint standing because joint standing is not metric transmission and supplies no boundary-controlled channel.

The manuscript establishes the third claim and the boundary no-channel constraint. It does not derive the full quantum probability law.

Definition 12.4 (Boundary-supplied channel). A boundary-supplied channel is a message-bearing map from sender-controlled inputs to receiver-accessible outputs whose dependence is supplied by a boundary selector, action, or standing transfer rather than licensed interior dynamics. Operationally, it requires declared same-domain sender interventions such that

$$\mathbb{P}(b \mid x = x_1, y) \neq \mathbb{P}(b \mid x = x_2, y)$$

for some x_1, x_2, y , with b accessible as a receiver record and with the intervention preserving the same target domain. Here the displayed probabilities are the receiver distributions under those interventions, not merely observational conditionals that might vary through common causes or selection. Preparation, receiver readout, and witness-channel identity remain fixed in the comparison.

Definition 12.5 (Boundary-independent interior channel). A boundary-independent channel is an interior intervention relation $\text{Chan} : X \rightarrow \Delta(Y)$ whose message dependence is generated by licensed interior dynamics and admitted transport paths, not by a boundary selector, boundary operator, or standing transfer.

Theorem 12.6 (No boundary-supplied superluminal channel). *Entanglement-style correlation is compatible with joint AMetric standing. The qualification and licensing relations alone supply no additional controllable communication channel, hence no additional superluminal channel. Any proposed sender-to-receiver dependence must be warranted by the original physical channel conditions; it cannot acquire that warrant merely from the standing or licensing of its records.*

Construction witness. A sender-controlled signal requires a warranted receiver-output difference under the fixed intervention comparison. By Theorem 6.2, licensing retains the original qualification, full use profile, and actual transition warrants; it introduces no further output law. If a richer

receiver-output answer is asserted from a description that forgets the information distinguishing that answer, the fibre criterion of Section 4 requires it to be constant on each fibre. A violation prevents that asserted descent. If the needed distinction or transition is already present in the original physical model, it is available on that model's warrant, not generated by licensing. Similarly, qualification of the sender record does not discharge the receiver's separate or joint original conditions. Treating that inference as its own warrant is the primitive self-authorization excluded in Theorem 4.6.

Thus a proposed selector, operator, or transport law is not rejected by its name: its actual original-domain warrant must be supplied. An admitted state change, interaction, or transport map defines an interior channel governed by the physical model. A change of preparation, receiver interface, or witness set defines a changed comparison and must be assessed there. In neither case does the qualification relation itself add the channel. This is precisely the no-added-channel consequence of Corollary 6.4, and in particular excludes a superluminal channel generated by licensing alone. No equality of quantum marginals is obtained from channel identity alone. $\square$

Remark 12.7 (Relation to quantum no-signaling). This theorem does not derive the quantum no-signaling equalities from Hilbert-space dynamics. Quantum no-signaling remains an interior formal result of the quantum regime. This also does not deny Bell's theorem or replace its locality assumptions. Bell-locality, parameter independence, and outcome independence are interior formal conditions on probability models. The theorem says only that the original qualification and licensing relations cannot themselves add a controllable signaling channel behind or beneath those formal results. Its proof uses conservation of the original use and channel warrants, not a blanket prohibition of lawful selectors or operators.

Remark 12.8 (Apparent instantaneousness). If joint compatibility is fixed AMetrically and later represented by spacelike-separated records, their correlation does not establish a transmitted boundary influence. A temporal ordering or an appearance of simultaneity belongs to the chosen interior representation. The licensing result supplies no preferred simultaneity relation; the physical model supplies the metric and dynamical account of the detections.

Part III

Consequences, limits, and refutation burden

13 Cascade compatibility with the realized physics arc

The L-metric boundary would be thin if it only blocked metric primitivity. Its positive function is that it licenses physical-interior roles derived in the realized-physics arc. This manuscript does not reprove that arc. The licensing theorem is independent of the full cascade; the cascade supplies examples of interior roles to which the theorem later applies.

Proposition 13.1 (Cascade compatibility). *For the realized-physics role constructions on their respective stated domains and hypotheses, compatibility with the L-metric interface requires their physical uses to satisfy the same-domain license. A role satisfying that license is an admissible interior use; promoting it into a primitive source of its own boundary standing is inadmissible. The name or downstream position of a role alone does not establish its construction or its license.*

Construction witness. Take a role construction with its source carrier, admitted witnesses, transformations, and continuation uses. Identify those data with the receiving physical domain and verify the licensing conditions there. The licensing theorem then permits the corresponding same-domain

interior use. If instead the role is used to create or revise the standing of its own prerequisites, it acquires boundary authority and the no-backflow result applies. This proves the compatibility claim at the licensing interface. Existence and coherence of the role construction remain the burdens discharged by the cited role theorem on its own hypotheses; the licensing theorem does not replace them. $\square$

13.1 Compact dependency summary

The dependency order may be displayed schematically as

$$\begin{aligned} \text{RealPers}_D &\longrightarrow \text{NonVacRec}_D \longrightarrow [B_D]_{\equiv B} \longrightarrow (\text{Adj}_D, N_D, \text{LCD}_D) \\ &\longrightarrow C_D \longrightarrow \text{BoundTrans}_D \longrightarrow \Phi_D \longrightarrow \Pi_D \longrightarrow G_D \longrightarrow M_D \longrightarrow \Omega_D \longrightarrow (T_D, C_D^{\text{caus}}). \end{aligned}$$

Each arrow records source dependency with the preceding role package and the additional hypotheses of that step; it is not an implication from the displayed predecessor alone. For example, R2 requires restorative continuation constraints, R3 adds boundary-fixed cross-locus continuation relevance, and R4 adds stable overlap and coherent quotient comparison. R5 constrains an admitted perturbation transport; it does not derive a finite numerical velocity. The exact receiving conditions and source theorem anchors are recorded in Appendix C. Here C_D in the middle denotes the quotient comparison carrier, while C_D^{caus} denotes the later causal role.

The synthesis theorem in R12 applies when the role families are jointly instantiated and cross-compatible [23, Theorem 6.1]. It does not infer their joint realization from the diagram. In particular, the partial comparison carrier of R4 need not be a real-valued metric, and the partial field-carrier assignment of R6 need not exist outside its transport-participating domains. These source domains remain fixed when the roles receive an L-metric interpretation.

Remark 13.2 (Why the cascade is not the proof). A reader need not accept every downstream realized-physics role to evaluate the licensing theorem. The theorem states that metric descriptors cannot be primitive boundary authority. The cascade shows where many physical descriptors naturally live if the realized-physics arc is adopted.

14 Target-typed GR/QM projection consequence

This section is deliberately modest. It is not a solution to quantum gravity and it does not define GR or QM from first principles. It states a boundary-order condition relevant to any account that treats GR-like and QM-like descriptions as interior projection regimes.

Principle 14.1 (Target-typed projection non-identity). *Within a non-degenerate target regime in which GR-like metric-geometric descriptions and QM-like compatibility/record descriptions are both licensed interior projection regimes, their coherence need not require primitive boundary-level identity. The relevant question is projection coherence on the target domain, not ontic fusion at the AMetric boundary.*

Construction witness. A projection regime may organize one family of interior descriptors: metric-geometric structure, record-compatibility structure, gauge-presentation structure, perturbation-transport structure, or statistical prediction structure. If two such regimes are downstream of L-metric licensing, neither is automatically primitive at the boundary. A demand for boundary-level fusion treats their interior roles as if they were competing grounds of admissibility. The correct question is whether their standing-preserving projections cohere on the target domain. This principle states a boundary-order condition any unification attempt must respect; it does not replace technical quantum-gravity work. $\square$

Remark 14.2 (Boundary-level overread of unification). A unification program may be valuable as interior mathematical physics even if it begins with GR metric geometry and QM operator/probability structure. The L-metric caution is only this: downstream interior formalisms should not be overread as primitive boundary ontologies merely because both are powerful.

15 Consequences and cautions: horizons, cosmology, constants

This section records controlled consequences. It intentionally avoids turning the L-metric theorem into a substitute for technical cosmology, quantum gravity, or numerical constant derivation.

15.1 Horizons and singularities

Proposition 15.1 (Metric horizon interiority). *A horizon is an interior metric construct. It may constrain metric description inside a licensed physical regime, but it cannot function as an admissibility boundary unless independently gated as such.*

Construction witness. A horizon is defined through the physical model’s metric, causal, or propagation structure. Licensing records its use under that model’s original conditions and does not supply those structures. A horizon condition may itself occur in an independently specified original physical question; its qualification role is then established by that question and its warrant. The mere existence or name of the horizon cannot confer this additional role. Without such an independent match it remains an interior constraint, not an additional source of admissibility authority. $\square$

Corollary 15.2 (Singularity caution). *A singularity indicates breakdown, incompleteness, or limit behavior of an interior metric representation. It does not automatically establish ontic terminality or AMetric admissibility failure.*

15.2 Cosmological transport

Cosmological inference often links early and late regimes by calibrated metric transports: sound horizons, luminosity distances, redshift-distance relations, and horizon-scale observables. The L-metric theorem does not reject those transports. It requires their standing-preserving transport structure to be explicit.

Proposition 15.3 (Cosmological transport audit). *Cross-regime cosmological comparison requires explicit target fixation, calibration transfer, metric comparison standards, and redescription-stable observable reconstruction. Without these, a disagreement may arise from transport or projection failure rather than from primitive metric ontology failure.*

Construction witness. A comparison between early and late regimes uses more than numbers. It requires a map between target fixations, calibrated observables, unit/normalization choices, and metric comparison standards. If those conditions are not explicit, elevating the disagreement to boundary-level metric failure would be metric backflow. The proper first step is a transport audit. $\square$

Remark 15.4 (Cosmology examples). Hubble-tension-style disagreements, sound-horizon calibration, luminosity-distance inference, and early/late projection divergence are natural test cases for this classification. Section 11.2 records one implemented transport-closure audit for the Hubble comparison. The present manuscript uses that result only as a case study in cross-regime metric transport: it does not adjudicate all cosmological models, reject SH0ES, or exclude new-physics extensions outside the tested basis [26, 33].

15.3 Constants, presentations, and ratios

The constants section fixes the legality conditions for future numerical extraction without claiming numerical derivations in the present manuscript.

Definition 15.5 (Dimensionful presentation). A dimensionful presentation is a numerical expression whose value depends on a unit, calibration, scale, or measurement convention. Examples include dimensional speeds, lengths, masses, energies, time intervals, and dimensional presentations of coupling or gravitational parameters.

Definition 15.6 (Dimensionless ratio candidate). A ratio candidate is a dimensionless relation that may survive changes of units or metric scale convention and may therefore be eligible for deeper tensor-level analysis, provided it is invariant under admissible redescription and tied to a role in the realized physical interior.

Theorem 15.7 (Dimensionful presentation interiority). *A dimensionful numerical presentation of a physical constant presupposes its unit/measurement regime. Its chart value alone cannot supply a unit-independent physical answer or primitive authority for the qualification of that regime.*

Construction witness. A dimensionful quantity has representations in unit-indexed output spaces. For a lawful change of units r , the same quantity is expressed by $F(rx) = \tau_r F(x)$, with the independently specified conversion τ_r . When that conversion changes the number, the raw number does not descend as a scalar on the physical quotient. The transported quantity can nevertheless be well defined and physically determined. Its lawful representation requires the original unit and target interpretation, so asserting the numeral cannot create those warrants. This is the typed descent test, not an assertion that every original qualification question excludes quantitative conditions. A physical condition already present in Q_D retains its original role. $\square$

Remark 15.8 (Unit convention caveat). The theorem concerns dimensionful numerical presentation, not the physical role of the constant in a theory. Setting $c = \hbar = 1$ does not erase physical structure; it changes presentation. A dimensionful physical quantity is compared using the admitted output transport of Section 10; different unit numbers alone do not constitute a physical difference. The role-bearing content must be recovered through invariant relations, dimensional analysis, and dimensionless combinations. Dimensionlessness alone likewise does not remove scheme or normalization dependence: the coupling source's raw chart value g may change to $f(g)$ while the represented physical trajectory is preserved [27, Appendix B].

Theorem 15.9 (Ratio candidacy theorem). *A dimensionless ratio may be a candidate for deeper AASC extraction only if it is invariant under admissible redescription, descends to the relevant quotient, preserves the witness ledger, and functions as tensor-level role content rather than skin-level numerical presentation.*

Construction witness. For the claimed invariant scalar readout, descent requires equality on the lawful representation fibres and the original witness and role conditions. Dimensionlessness alone proves none of these conditions: a dimensionless chart coordinate may still change under a scheme map. Such a coordinate can represent physical content through a proved output transport, but its raw value is not the invariant scalar claimed here. Conversely an invariant number without the required physical interpretation supplies no extraction of that target. A ratio satisfying the descent and role conditions is eligible for further analysis; its actual value still requires a determining argument. No classification of a raw component as wholly skin follows merely from one varying coordinate. $\square$

Corollary 15.10 (Numerical readiness criterion). *For an invariant scalar numerical target, readiness requires a fixed physical role and witness structure, descent under the relevant scheme, coordinate, gauge, normalization, and unit transformations, and a separate determination argument beyond its being fitted. A dimensionful target can instead be formulated in transported output spaces with an independently fixed unit convention or scale anchor. Readiness in either form is a condition for extraction, not a derivation of the value.*

Remark 15.11 (Numerical restraint). The legality conditions above do not predict constants. They prevent treating fitted numbers or untransported chart coordinates as self-authorizing physical answers. Future numerical work must supply the original role, lawful scalar descent or typed output transport, and a determining argument, including any required scale anchor [27, 28, 7].

16 Objections, counterexamples, and proof obligations

The central theorem is answerable to ordinary mathematical and scientific criticism. Its claimed physical application needs the original-domain interpretation, complete use coverage, and source-to-target conditions displayed in the proof. Kernel necessity does not excuse a missing receiving hypothesis. Nor does a list of failure names establish that a proposed alternative fails.

16.1 What an objection can establish

A criticism can identify a false premise, an invalid inference, an unproved interpretation or coverage claim, or a counterinstance to a stated conclusion. These are different outcomes. A gap in the offered proof does not by itself show the theorem false, and a missing certificate does not show the physical qualification negative. Conversely, identifying a precise unsupported inference is a legitimate objection without first constructing a new physical theory.

A challenge to Kernel necessity must address its actual-incidence argument: why retaining the same realized subject, relation, joint conditions, and original occurrence would not require the work reconstructed in Lemma 2.4. A challenge to this paper’s downstream theorem can instead address its own receiving construction. For example, a required use omitted from U_D , an unretained operation domain, an unsafe declared continuation, or an unsupported descriptor transport defeats the corresponding application claim without denying Kernel governance.

Corollary 16.1 (Audit-local and theorem-level claims). *Failure of a physics audit affects the claims depending on that audit. It is not by itself a counterexample to the independently proved licensing construction. If it exposes an interpretation or preservation condition also used in an application of that construction, the affected application must be corrected.*

16.2 Counterinstance and proof-gap tests

Theorem 16.2 (Faithful same-task counterinstance criterion). *A counterinstance to the fixed-task licensing theorem must retain the independently interpreted original task, satisfy the theorem’s actual hypotheses, and violate a stated conclusion. It need not avoid the words metric, selector, operator, parameter, or constructor. A demonstrated failure of requirement coverage, descriptor transport, original-incidence warrant, or an inference in the proof instead defeats the corresponding justification or claimed application, even when no counterinstance to the correctly conditional theorem is supplied.*

Construction witness. The first statement is the logical test for a conditional theorem. Section 6 makes its nontrivial receiving obligations inspectable: the original qualification, complete question

family, full promised repertoire, local and edge conditions, readiness, and redescription transport. Each proved conclusion uses specified obligations. If one is absent or false, that application of the conditional result is unestablished; if all hold and the conclusion fails, the conditional theorem is false. Mechanism names add no premise to this test. The graph, fibre, and path witnesses in Section 3 locate semantic violations, while an unproved interpretation remains a proof obligation rather than a negative verdict. $\square$

Remark 16.3 (Exhaustion is itself accountable). A proposed new implementation does not evade an original semantic requirement by changing its notation. Equally, an alleged exhaustion cannot discard a relevant original question because it does not fit its taxonomy. The complete profile must retain that question or prove its recovery. Demonstrated misclassification or omitted coverage is sufficient to challenge the corresponding proof; an unnamed extra category is not required.

16.3 Objection matrix

Objection	Response and actual proof burden
Metric structure is already physically real.	Kernel governance concerns that actual incidence, including unobserved geometry. Its description answers to the original conditions; licensing is not physical creation.
A false metric claim is still meaningful.	Yes. Weak and target-fixed interpretation do not imply truth. Lemma 6.1 concerns the admissible original use and its own qualification.
No preferred point does not exclude a metric.	Correct. The AMetric application uses the actual retained structure and lawful action. An unmarked cycle retains adjacency and its graph metric; equality-only input is a separate hypothesis.
An operator or experiment establishes a new result.	Its independently warranted domain and original result incidence can establish qualification. The exclusion concerns an unsupported first license, not every process or constructor.
Evidence establishes standing.	Evidence can warrant an original claim or a new use through its actual incidence and transfer conditions. It cannot replace an unchanged occurrence's false original conditions with later success.
A positive bit licenses every subsequent use.	No. Required domains, joint conditions, and future questions remain in the complete profile. Exact robust transport also needs target-successor coverage.
Joint standing produces separated quantum records.	The audit uses the admitted model's record projections and witness table. Licensing alone produces neither that map nor the quantum probability law.
A finite audit certifies universal closure.	A finite check can test its instances. Universal closure here rests on original semantic coverage and the graph, fibre, and continuation proofs.

Objection	Response and actual proof burden
A positive mechanism generating metricity is missing.	The central result licenses metric uses and preserves non-generation. A proposed physical mechanism has its own original domain and dynamics; this paper does not derive them.

These responses delimit the theorem without insulating its scientific application. The strongest justified result is the constructed licensing relation and its exact preservation and obstruction statements. A universal physical realization, complete empirical theory, or guaranteed acceptance by every reviewer is not a consequence.

17 Dependency map of main results

The current foundational imports and the historical physics applications play different proof roles. The source editions are explicit in the bibliography; their frozen internal imports retain their original identities. In particular, current Kernel necessity is not silently inserted into an older source's dependency graph.

- R1. Actual-incidence necessity.** Kernel Proposition 2.4 and Theorem 3.1 derive the functional commitments and roles from actual non-degenerate determination, including unobserved incidences.
- R2. Use-specific qualification.** Kernel Definition 2.5 and Proposition 2.6 provide original qualification and its faithful Boolean report. The candidate inscription and the physical claim made with it remain different uses.
- R3. AMetric restrictions.** The actual incidence, eligible-stabilizer, retained-data-fibre, and grounded-license conditions support constructed Axiom 6, Theorem 7.5. Equality-only input is not inferred from absence of a preferred atom.
- R4. Complete semantic coverage.** Structure Lemma 8.4 and Exhaustion Theorem 3.2 retain each independently specified original question. A smaller realization needs its own coverage and exact-decoding proof.
- R5. Exhaustion with witnesses.** Structure Theorem 8.6 and Exhaustion Theorem 5.9 supply original-result, information, continuation, and support alternatives. Mechanism names follow the witness.
- R6. Licensing construction.** Structure Theorem 6.3 and Exhaustion Theorem 5.3 instantiate the greatest safe-use construction on C_D, A_D, R_D, W_D . Theorem 6.2 projects its records and proves (L1)–(L5).
- R7. Non-generation and no-backflow.** Original-incidence dependence and AMetric's fibre and grounding proofs preserve the licensing endpoint. Historical retention is distinguished from negative endpoint absorption.
- R8. ATS roles.** ATS Theorems 3.1 and 4.2–4.4 classify the context, content, and realization effects of specified variations, with refinement of mixed raw components. ATS is downstream of semantic exhaustion.

- R9. Exact transport.** Unus Theorem 5.3 requires local-validity equivalence and successor coverage in both directions for exact safety-operator transport. Target coverage and other readout equations remain explicit.
- R10. Joint use.** Exhaustion Theorem 4.8 retains joint constructor domains; Unus Theorems 6.1–6.2 retain mixed continuation paths. Separate positive components alone do not prove joint qualification.
- R11. Locality.** R2/R3 supply their pre-metric roles under their actual source hypotheses. R5’s path-bounded transport is distinguished from a quantitative finite-speed law. Unus Theorem 6.4 supplies the exact unchanged-contract extension criterion.
- R12. Fixation-local comparison.** Lorentzian and cross-fixation uses require their original relata and warranted comparison or transport. No existence of a metric follows from a license alone.
- R13. Quotient and ZPE audit.** Invariant scalar answers factor through the specified quotient; typed outputs use coherent transport. The compensated homogeneous vacuum shift defeats an absolute matter-sector scalar prediction while preserving the specified gravitational combination.
- R14. SM audit.** Structural-role claims are kept separate from empirical parameter completion and from the frozen source’s conditional lepton and neutrino-sector results.
- R15. Cosmological transport.** Anchor, witness, calibration, estimator-role, and error/likelihood conditions belong to the actual comparison. A fitted diagnostic is not universal semantic closure.
- R16. No boundary channel.** The admitted Bell witness, interventions, and record projections are supplied by the physical model. Licensing adds no message mechanism or quantum dynamics.
- R17. Guarded cascade.** Each R1–R12 dependency retains its source hypotheses and joint-compatibility burden. The cascade is an application, not a premise of the licensing theorem.
- R18. Remaining physical questions.** GR/QM dynamics, horizons, constants, ratios, and model selection retain their own target-specific existence, incidence, and prediction burdens.

Remark 17.1 (Strength of the endpoint). The relation is complete for its independently specified original metric-use task and contains every sound region of that contract. This does not prove nonempty realization of an inconsistent contract, existence or global uniqueness of a minimal L-metric interface, uniqueness of a metric or physical occupant, or closure of every future physical question. These are distinct statements with distinct proof obligations.

18 Conclusion: metric physics as licensed description

The L-metric boundary names a locally minimal interface where physical metric description is available. The result concerns the conditions of that use, rather than a mechanism creating geometry. Actual determinate physical incidences necessarily incur Kernel governance; their particular metric relations and qualifications remain fixed by their original physical conditions.

The central construction retains the complete original metric-use question, its answer and application domains, joint conditions, and declared continuations. Its greatest safe region supplies a canonical licensing relation. Theorem 6.2 verifies domain qualification, redescription, boundary-work preservation, continuation compatibility, and non-generativity separately. Weakly interpretable candidates remain available before their target-specific standing is established. An actual false report remains a determinate inscription without acquiring qualification for its false claim.

The endpoint is

$$\text{admissible interpreted metric use} \implies (a, m) \in \text{Lic}_D \implies L_D(a) = 1.$$

The original incidence grounds the right-hand qualification; the license records its admissible metric articulation. This implication neither constructs a physical metric nor guarantees an admissible metric use for every proposed domain. The boundary licenses description and does not generate metric structure, standing, or ontology.

The AMetric constraints identify the exact failed moves: an unsupported answer to the original question, a selector lacking its eligible fixed point, an answer requiring a distinction lost by the retained input, or a license family with no grounded entrance. Original metric conditions, lawful selectors, new evidence, and sound constructors remain admissible on their own warrants. A later success can establish a new qualified use or reveal an old truth; it cannot substitute new original data for an unchanged failed occurrence.

The physics audits carry these distinctions into Lorentzian comparison, the compensated vacuum-energy quotient, structural and numerical SM claims, cosmological transport, and quantum record/channel questions. They retain their own physical hypotheses and supply no premise of the foundational licensing proof. Metric physics remains a substantive account of its admitted physical subject matter. Its licensed descriptions cannot replace the original conditions on which their own use depends.

A Glossary of manuscript terms

Term	Use in this manuscript
Background same-scope	Preservation of acts, evaluated fragment, redescription, identity, continuation, and boundary-place type. The disputed role and a competing standing map remain open to original-incidence testing.
Standing-preserving same-scope	Background same-scope plus preservation of the standing map on evaluated acts.
Admissibility standing	The faithful Boolean report of the complete original use's qualification $Q_D(a)$ on the evaluated fragment. It does not replace the use's substantive conditions.
Empirical warrant	Evidential support for use, trust, comparison, or continuation of a model inside scientific practice. Not identical to primitive boundary standing.
AMetric boundary	The original-incidence, retained-data, and grounding constraints excluding a description as the primitive source of its own missing target qualification.
L-metric boundary	The minimal downstream projection interface at which metric descriptors may become physically interpretable.

Term	Use in this manuscript
Projection order $\preceq_D$	Dependency order among projection interfaces; not temporal order.
Metric descriptor	A distance, topology/support, causal, dynamical, observational, or constant-presentation descriptor requiring metric-interior articulation.
Candidate application class	The syntactic class $X_D(m)$ on which a descriptor may be written before physical interpretability has been established.
Licensing	The constrained relation $\text{Lic}_D \subseteq \text{Int}_D^+ \times \mathcal{M}_D$ allowing standing-positive content to receive metric articulation without changing boundary standing.
Boundary authority	An alleged effect on classification, continuation, admission, or outcome selection. Its warrant must be established; primitive self-authorization substitutes an asserted license for original qualification.
Metric backflow	Retro-authorizing boundary standing by appeal to later interior metric success or structure.
Quotient descent	Factorization of a scalar answer with fixed meaning through the declared quotient. Dimensional or covariant outputs instead use independently justified coherent output transport.
Standing-preserving representation	Interior representation that preserves target fixation, redescription discipline, and same-domain continuation.
Boundary-supplied channel	A message-bearing intervention relation whose admissibility depends on boundary selector/operator/standing transfer.
Dimensionful presentation	Unit-, calibration-, scale-, or measurement-dependent numerical expression of a physical parameter or constant.
Ratio candidate	A dimensionless relation considered for deeper extraction only if redescription-invariant, quotient-stable, witness-preserving, and tensor-role bearing.

B Normal form of the L-metric argument

The source-faithful logical spine is as follows.

- N1.** Begin with actual determinate incidences and the Kernel necessity argument; do not infer physical realization from an uninterpreted symbol.
- N2.** Fix the original bearer, occurrence, complete use question, and substantive condition Q_D before assessing a proposed standing map.
- N3.** Retain weak candidate interpretation. Establish the map ι_m to the particular original use separately from that use's qualification.

- N4.** Specify all original required answers, including definedness, joint domains, boundary work, and promised continuations. Construct their complete semantic profile.
- N5.** Apply the actual AMetric incidence, stabilizer, fibre, and grounding tests. Preserve richer original relations, lawful selections, and independently warranted constructions.
- N6.** Define the local L-metric interface weakly. Its positive realization and minimality require their own witnesses; neither follows merely from defining a projection order.
- N7.** Specify the candidate metric-use carrier C_D , original local conditions A_D , declared repertoire R_D , and original edge warrants W_D independently of Lic_D .
- N8.** Construct the greatest safe region S_D and prove its fixed-point and greatestness equations by finite-path reasoning over arbitrary declared branching.
- N9.** Project its complete original-use records to Lic_D . Prove (L1)–(L5), including descriptor-family closure and full original-profile transport under every required redescription.
- N10.** Separate present qualification, robust reuse, and joint use. A transport of the whole safety contract must preserve local validity and cover target successors; equal positive bits alone are insufficient.
- N11.** Retain historical non-substitution and exact fibre/grounding obstructions. Do not convert them into a prohibition on new successful objects or discovery of an original truth.
- N12.** Apply the licensing result to physical models only with their original source hypotheses. Scalar quotient descent uses fixed output meaning; dimensional and covariant quantities use coherent output transport.

The construction licenses metric description without generating its metric structure, standing, ontology, dynamics, or numerical parameter values. Its scientific applications remain accountable to the original physical conditions and the stated source-to-target matches.

C Realized-physics cascade table

This table is imported dependency structure, not a proof newly supplied in this manuscript.

Every row retains the fixed admissible realized-persistence domain of its source. In addition to the earlier role package, the following receiving hypotheses must be checked before a role is used. Source references are to the frozen R1–R12 texts; their numbered theorem statements control the abbreviated dependency arrows in Section 13.

Source result	Additional receiving condition and exact conclusion
R1, Theorem 4.14	Minimal realization and actual standing-bearing persistence, with the admissible reconstruction family of Section 4, imply non-vacuous reconstruction. Persistence capability alone does not suffice.
R2, Theorem 7.1	Realized persistence with restorative continuation constraints implies a bounded standing-localization class. Corollary 7.2 obtains those constraints from standing-relevant perturbation and lawful restoration capacity. Boundedness is role-boundedness, not finite spatial diameter.

Source result	Additional receiving condition and exact conclusion
R3, Theorems 7.1 and 9.3	Bounded loci plus boundary-fixed, standing-relevant cross-locus continuation induce adjacency and continuation-neighborhoods. Admissibly composable adjacency paths induce a local continuation domain; locality alone supplies no edge.
R4, Theorem 11.1	Stable overlap witnesses, path-coherent transport, and quotient-stable comparison induce $C_D = \text{CompWit}_D / \sim_D$ on the admitted comparison domain. Multiple inequivalent witnesses give a relation; a single-valued representation requires a fixed comparison class. Real-valued metric axioms remain additional.
R5, Theorem 9.1	A standing-relevant perturbation whose transport is admitted must follow continuation-neighborhood paths with boundary-preserving witnesses and coherent comparison-class composition. No numerical speed, temporal duration, or relativistic cone follows from this role condition alone.
R6, Theorems 8.1 and 9.1	Localized persistence retained under bounded perturbation transport induces carrier profiles cohering under admitted restriction, overlap, and transport. The assignment is partial on participating domains; no global field substrate is supplied by default.
R7, Theorem 6.1	A localized carrier-mediated configuration with stable localization and retained identity under admitted continuation induces a particle-role class. The existence of field carriers alone does not ensure a stable particle class.
R8, Theorem 7.1	Boundary-fixed multiplicity of presentations preserving the same carrier/particle roles induces gauge-role equivalence and compatible transitions. A particular Lie group requires later specialization.
R9, Theorem 7.1	An admitted identity-retaining deformation family induces a quotient-stable mass-response profile; its transport-state restriction gives inertia. Numerical masses and gravitational coupling are not supplied by this result.
R10, Theorem 7.1	Minimal concrete persistence with no localized particle-class excitations, while the earlier carrier and response roles remain, induces the non-null vacuum baseline. Its role does not fix an absolute vacuum-energy number.
R11, Theorems 7.2 and 8.1	Irreversible standing-bearing continuation, record fixation, and identity retention induce the temporal role. Dependence on an earlier fixed witness supplies the causal role. Temporal order may be partial and does not itself give a relativistic causal metric.
R12, Theorem 6.1	Joint instantiation and cross-compatibility of all preceding role families give the coherent, role-closed, operationally minimal interior architecture. The theorem does not supply missing instantiations.

The source chain is R1–R12 [12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23]. Its role conclusions receive the following L-metric classification only on those source conditions and the receiving license.

Role	L-metric status
RealPers_D	Realized persistence; non-null physical interior precondition, not metricity itself.
NonVacRec_D	Non-vacuous record structure; gives persistence a retained witness role.
$[B_D]_{\equiv B}$	Boundary class under standing-preserving equivalence; not a generative surface.
$(\text{Adj}_D, N_D, \text{LCD}_D)$	Adjacency, neighborhood, and local continuation domain roles; pre-metric locality structure.
C_D	Quotient comparison carrier; the first natural site for metric comparison roles.
BoundTrans_D	Perturbation transport constrained to admitted continuation-neighborhood paths; no numerical propagation speed follows.
Φ_D	Field-carrier role; persistence support profile, not primitive substrate.
Π_D	Particle/localized persistence class role; not primitive point-object role.
G_D	Gauge role from boundary-trace fixation and presentation equivalence.
M_D	Mass/inertia role as persistence compression or deformation-response profile.
Ω_D	Vacuum role as minimal concrete persistence regime, not empty spacetime.
(T_D, C_D^{caus})	Time and causality from irreversible continuation ordering; not primitive boundary time.

D Detailed Standard Model structural/parametric audit table

The main text uses the Standard Model case only for the structural/parametric typing distinction. The source’s tuple, its receiving restrictions, and the detailed role table are retained here for auditability. Their inclusion does not make the companion Standard Model derivation a premise of the L-metric theorem.

The frozen Standard Model support paper explicitly separates these levels (Criterion 1.6, Definition 1.7, and Theorem 12.2). Its structural target is the source’s tuple

$$\Sigma_{\text{SM}} = (G_{\text{SM}}, R_{\text{SM}}, H_{\text{br/fix}}, O_Y, Q_{\text{SM}}, A_{\text{anom}}, V_{\text{SM}}, M_{\text{slot}}, \Theta_{\text{mix}}, N_g) / \simeq_{\text{Adm}},$$

where the entries record gauge-product structure, representation-response roles, Higgs/bridge fixation, Yukawa-realizability and coupling-slot topology, charge lattice, anomaly-neutral survivor structure, vacuum compatibility, mass-response slots, mixing/phase slots, and the family-cardinality envelope record. The same support paper states that gauge-coupling numerical values, Yukawa eigenvalues, quark masses, CKM/PMNS coordinates, Higgs-potential data, weak-scale values, and absolute scale anchors remain downstream unless later modules supply the relevant witness, exactness certificate, reconstruction rule, or zero-anchor derivation [32].

Here N_g retains the source’s structural envelope status: changing the family count changes representation and mixing roles, so it is not merely a fitted mass or coupling coordinate. The source’s placement of $N_g = 3$ in that envelope is not imported as a derivation of the integer by the charge/anomaly mechanism. Likewise, the neutrino operator slot keeps the source’s declared sector status; no Dirac, Majorana, Weinberg, or sterile-sector choice follows from metric licensing. These are receiving-type restrictions on the example, not additional premises of the licensing theorem.

Standard Model object	L-metric/AASC role
$SU(3)_c \times SU(2)_L \times U(1)_Y$	Licensed structural gauge-product presentation, after admissible role-carrier and survivor classification.
Standard Model Lagrangian presentation	Compact interior formal presentation, not a primitive boundary premise.
Chiral representation table	Standing-preserving representation-response skeleton.
Charge lattice	Tensor-level structural ledger, modulo admissible normalization/sign redescription.
Anomaly cancellation	Admissibility obstruction and survivor condition; not a fitted metric number.
Higgs-like bridge/fixation carrier	Structural bridge for order-to-deformation mediation; not numerical Higgs-sector completion.
Yukawa-realizability ledger O_Y	Operator-availability and coupling-slot topology over the bridge carrier.
Numerical Yukawa matrices	Downstream parameter-sector objects, not structural proof inputs.
Gauge-coupling values	Parameter/moduli targets unless a later exactness, boundary, dynamical, or quotient certificate is supplied.
Charged-lepton one-anchor readout	Conditional metric display using one scale anchor in the frozen source; its protected residual-localization result leaves exact pole closure open.
Zero-anchor absolute scale	Explicitly open in the frozen source unless a scale-carrier theorem supplies the missing anchor.

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